

ON TESTING FOR HIGH-DIMENSIONAL WHITE NOISE

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Testing for white noise is a classical yet important problem in statistics, especially for diagnostic checks in time series modeling and linear regression. For high-dimensional time series in the sense that the dimension p is large in relation to the sample size T , the popular omnibus tests including the multivariate Hosking and Li-McLeod tests are extremely conservative, leading to substantial power loss. To develop more relevant tests for high-dimensional cases, we propose a portmanteau-type test statistic which is the sum of squared singular values of the first q lagged sample autocovariance matrices. It, therefore, encapsulates all the serial correlations (upto the time lag q) within and across all component series. Using the tools from random matrix theory, we derive the normal limiting distributions, as both p and T diverge to infinity, for this test statistic. As the actual implementation of the test requires the knowledge of three characteristic constants of the population cross-sectional covariance matrix and the value of the fourth moment of the standardized innovations, non trivial estimations are proposed for these parameters and their integration leads to a practically usable test. Extensive simulation confirms the excellent finite-sample performance of the new test with accurate size and satisfactory power for a large range of finite (p, T) combinations, therefore ensuring wide applicability in practice. In particular, the new tests are consistently superior to the traditional Hosking and Li-McLeod tests.

1. Introduction. Testing for white noise is an important problem in statistics. It is indispensable in diagnostic checking for linear regression and linear time series modeling in particular. The surge of recent interests in modeling high-dimensional time series adds a further challenge: diagnostic checking demands the testing for high-dimensional white noise in the sense that the dimension of time series is comparable to or even greater than the sample size (i.e., the observed length of the time series). One prominent example showing the need for diagnostic checking in high-dimensional time series concerns the vector autoregressive model, which has a large literature. When dimension is large, most existing works regularize the fitted models by Lasso (5, 16, 18, 30), Dantzig penalization (15), banded autocovariances (6), or banded autocoefficient matrices (12). However, none of them have developed any residual-based diagnostic tools.

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Another popular approach is to represent high-dimensional time series by lower-dimensional factors. See for example, (33–35), (10, 11), (1), (22) and (8). Again, there is a pertinent need to develop appropriate tools for checking the validity of the fitted factor models through careful examination of the residuals.

There are several well-established white noise tests for univariate time series (23). Some of them have been extended for testing vector time series (17, 24, 27). However, these methods are designed for the cases where the dimension of the time series is small or relatively small compared to the sample size. For the purpose of model diagnostic checking, the so-called omnibus tests are often adopted which are designed to detect any forms of departure from white noise. The celebrated Box-Pierce portmanteau test and its variations are the most popular omnibus tests. The fact that the Box-Pierce test and its variations are asymptotically distribution-free and χ^2 -distributed under the null hypothesis makes them particularly easy to use in practice. However, it is well known in the literature that the slow convergence to their asymptotic null distributions is particularly pronounced in multivariate cases. On the other hand, testing for high-dimensional time series is still in an infancy stage. To our best knowledge, the only available methods are (9) and (36).

To appreciate the challenge in testing for a high-dimensional white noise, we refer to an example reported in Section 3.1 below where say, we have to check the residuals from a fitted multivariate volatility for a portfolio containing $p = 50$ stocks using their daily returns over a period of one semester. The length of the returns time series is then approximately $T = 100$. Table 1 shows that the two variants of the multivariate portmanteau test, namely the Hosking and Li-McLeod tests, all have actual sizes around 0.1%, instead of the nominal level of 5%. These omnibus tests are thus extremely conservative and they will not be able to detect an eventual misfitting of the volatility model.

The above example illustrates the following fact, which is now better understood: many popular tools in multivariate statistics are severely challenged by the emergence of high-dimensional data, and they need to be *re-examined* or *corrected*. Recent advances in high-dimensional statistics demonstrate that feasible and quality solutions to these high-dimensional challenges can be obtained by exploiting tools of random matrix theory via a precise spectral analysis of large sample covariance or sample autocovariance matrices. For a review on such progress, we refer to (20), (28) and monograph (38). In particular, asymptotic results found in this context using random matrix theory exhibit fast convergence rates, and hence provide satisfactory finite sample approximation for data analysis.

This paper proposes a new method for testing high-dimensional white noise. The test statistic encapsulates the serial correlations within and across all component series. Precisely, the statistic is the sum of the squared singular values of several lagged sample autocovariance matrices. Using random matrix theory, asymptotic normality for the test statistics is established under the Marčenko-Pastur asymptotic regime where p and T are large and comparable. Next, original methods are proposed for estimation of a few parameters in the limiting distribution in order to get a fully implementable version of the test. Extensive simulation demonstrates excellent behavior of the proposed tests for a wide array of combinations of (p, T) , with accurate size and satisfactory power. In this paper, we also explore the reasons why popular the multivariate Hosking and Li-McLeod tests are no longer reliable when the dimension is large in relation to

the sample size.

The rest of the paper is organized as follows. Section 2 presents the main contributions of the paper. A new high-dimensional test for white noise is introduced, its asymptotic null distributions established and a practically usable version of the test defined. Section 3 reports some extensive Monte-Carlo experiments which assess the finite sample behavior of the tests. Whenever possible, comparison is made with the popular Hosking and Li-McLeod tests, Numerical evidence also indicates that the new test is more powerful than that of (9). Section 4 collects all the technical proofs.

2. A test for high-dimensional white noise. Let $\mathbf{x}_1, \dots, \mathbf{x}_T$ be observations from a $p \times 1$ complex-valued linear process of the form

$$\mathbf{x}_t = \sum_{l \geq 0} A_l \mathbf{z}_{t-l},$$

where A_l are $p \times p$ coefficient matrices, $\{\mathbf{z}_t\}$ is a sequence of independent $p \times 1$ random vectors with independent components, $\mathbb{E}z_{it} = 0$, $\mathbb{E}|z_{it}|^2 = 1$ and $\mathbb{E}|z_{it}|^4 = \nu_4 < \infty$. Note that z_{it} denotes the i -th component of $\{\mathbf{z}_t\}$. Hence $\mathbb{E}\mathbf{x}_t = \mathbf{0}$, and $\Sigma_\tau \equiv \text{Cov}(\mathbf{x}_{t+\tau}, \mathbf{x}_t)$ depends on τ only. Note that $\Sigma_0 = \text{var}(\mathbf{x}_t)$ is the population covariance matrix of the time series. The goal is to test if null hypothesis

$$(2.1) \quad H_0 : \mathbf{x}_t = \Sigma_0^{1/2} \mathbf{z}_t$$

holds or not, where Σ_0 is unknown. This in fact tests the independence instead of linear independence (i.e. $\Sigma_\tau = 0$ for all $\tau \neq 0$), which is however a common practice in the literature of white noise tests. Throughout the paper, the complex adjoint of a matrix (or vector) A is denoted by A^* . For $\tau \geq 1$, let $\widehat{\Sigma}_\tau$ be the *lag τ sample autocovariance matrix*

$$\widehat{\Sigma}_\tau = \frac{1}{T} \sum_{t=1}^T \mathbf{x}_t \mathbf{x}_{t-\tau}^*,$$

where by convention $\mathbf{x}_t = \mathbf{x}_{T+t}$ when $t \leq 0$. Under the null hypothesis, $\mathbb{E}(\widehat{\Sigma}_\tau) = 0$ for $\tau \neq 0$, and a natural test statistic is the sum of squared singular values of the first q lagged sample autocovariance matrices:

$$G_q = \sum_{\tau=1}^q \text{Tr}(\widehat{\Sigma}_\tau^* \widehat{\Sigma}_\tau) = \sum_{\tau=1}^q \sum_j \alpha_{\tau,j}^2,$$

where $\{\alpha_{\tau,j}\}$ are the singular values of $\widehat{\Sigma}_\tau$, and Tr denotes the trace operation for square matrices. We reject null hypothesis H_0 for large values of G_q .

Notice that the setting here allows complex-valued observations: this is important for applications in areas such as signal processing where signal time series are usually complex-valued. However, for the sake of presentation, we mostly focus on the real-valued case in the subsequent sections. Directions on how the tests can be extended to accommodate complex-valued observations will be given in the last Section 2.4.

2.1. *High dimensional asymptotics.* We adopt the so-called *Marčenko-Pastur regime* for asymptotic analysis, i.e. we assume $c_p = p/T \rightarrow c > 0$ when $p, T \rightarrow \infty$. This asymptotic framework has been widely employed in the literature on high-dimensional statistics, see, (20), (28), also monograph (38) and the references within. Most the results in this area concern sample covariance matrices only. However our test statistic G_q is based on the sample autocovariance matrices, which is much less studied; see (26) and (7).

As a main contribution of the paper, we characterize the asymptotic distribution of G_q in this high-dimensional setting when the observations are real-valued. We introduce the following limits whenever they exist: for $\ell \geq 1$,

$$(2.2) \quad s_\ell = \lim_{p \rightarrow \infty} \frac{1}{p} \text{Tr}(\Sigma_0^\ell), \quad s_{d,\ell} = \lim_{p \rightarrow \infty} \frac{1}{p} \text{Tr}(\text{diag}^\ell(\Sigma_0)),$$

where $\text{diag}(A)$ denotes the diagonal matrix consisting of the main diagonal elements of A .

THEOREM 2.1. *Let $q \geq 1$ be a fixed integer, and the following assertions hold.*

1. $\{\mathbf{z}_t\}$ is a sequence of real-valued independent $p \times 1$ random vectors with independent components $\mathbf{z}_t = (z_{it})$ satisfying $\mathbb{E}z_{it} = 0$, $\mathbb{E}z_{it}^2 = 1$ and $\mathbb{E}z_{it}^4 = \nu_4 < \infty$;
2. $\{\Sigma_0\}$ is a sequence of $p \times p$ semi-positive definite matrices with bounded spectral norm such that the limits $\{s_1, s_2\}$ and $\{s_{d,2}\}$ exist;
3. (Marčenko-Pastur regime). The dimension p and the sample size T grow to infinity in a related way such that $c_p := p/T \rightarrow c > 0$.

Then when $\mathbf{x}_t = \Sigma_0^{1/2} \mathbf{z}_t$, the limiting distribution of the test statistic G_q is

$$(2.3) \quad G_q - qTc_p^2s_1^2 \xrightarrow{d} \mathcal{N}(0, \sigma^2(c)),$$

where

$$(2.4) \quad \sigma^2(c) = 2qc^2s_2^2 + 4q^2c^3(\nu_4 - 3)s_1^2s_{d,2} + 8q^2c^3s_1^2s_2.$$

The proof of this theorem is given in Section 4.

Let Z_α be the upper- α quantile of the standard normal distribution at level α . Based on Theorem 2.1, we obtain a procedure for testing the null hypothesis in (2.1) as follows.

$$(2.5) \quad \text{Reject } H_0 \text{ if } \{G_q - qTc_p^2s_1^2 > Z_\alpha\sigma(c)\}.$$

The illustration in Section 3 indicates that the test above is much more powerful than some classical alternatives, especially when the dimension p is growing linearly with the sample size T . The power of this test is gained from gathering together the serial correlations from the first q lags within and across all p component series; see the definition of G_q . Also note that the asymptotic mean of G_q is $qTc_p^2s_1^2$, which grows linearly with T (and p), while its asymptotic variance $\sigma^2(c)$ is a constant. This implies that even for moderately large T , departure from white noise in the first q lags of autocovariance matrix is likely to result in a large and different mean,

which will be a large multiple standard deviation away from $qTc_p^2s_1^2$ since the standard deviation $\sigma(c)$ is constant.

However the test G_q in (2.5) is not yet practically usable as it depends on (i) three characteristic constants of the (population) cross-sectional covariance matrix Σ_0 , namely s_1 , s_2 and $s_{d,2}$; and (ii) the fourth moment ν_4 of the innovations $\{\mathbf{z}_t\}$. These issues are addressed below.

2.2. Estimation of the covariance characteristics s_1 , s_2 and $s_{d,2}$. If the cross-sectional covariance matrix Σ_0 is known, deterministically consistent estimates of these characteristics are readily calculated from Σ_0 . By substituting these estimates in the asymptotic variance $\sigma^2(c)$ and the centering term $qTc_p^2s_1^2$, the resulting asymptotic distribution of G_q remains unchanged and the test (2.5) still applies.

However, the population covariance matrix Σ_0 is in general unknown and the situation becomes challenging as estimating a general Σ_0 is somehow out of reach without specific assumptions on its structure. Luckily, as observed previously, we only need “good” estimates of the three characteristics. First of all, in the setting of Theorem 2.1 and under the null, it is not difficult to find *consistent* estimators for these characteristics, thus a consistent estimator of the limiting variance $\sigma^2(c)$. The situation is much more intricate for the centering term $qTc_p^2s_1^2$. Suppose $\hat{s}_1^2$ is a consistent estimator of s_1^2 . Plugging it into the centering term leads to

$$(2.6) \quad G_1 - qTc_p^2\hat{s}_1^2 = \{G_1 - qTc_p^2s_1^2\} + qTc_p^2\{s_1^2 - \hat{s}_1^2\}.$$

Because of the multiplication by T here, the asymptotic distribution would remain the same only if the estimation error $\{\hat{s}_1^2 - s_1^2\}$ is of order $o_p(1/T)$. This is however not the case and in general the error is exactly of the order $O_p(1/T)$ and $T\{\hat{s}_1^2 - s_1^2\}$ converges to another normal distribution.

Our method is as follows. First we establish the asymptotic distribution of $T\{\hat{s}_1^2 - s_1^2\}$ for a natural estimator $\hat{s}_1^2$. Because the correlation between the statistic G_q and $\hat{s}_1^2$ is difficult to analyze, we thus proceed as follows:

1. Split the whole data set into two halves;
2. Use the two halves to calculate G_q and $\hat{s}_1^2$, respectively;
3. These two estimators being independent of each other and both asymptotically normal, the decomposition (2.6) readily gives the required asymptotic normal distribution.

We consider the sample covariance matrix $\widehat{\Sigma}_0 = \frac{1}{T} \sum_{t=1}^T \mathbf{x}_t \mathbf{x}_t^*$ and define the natural estimators of s_1 and s_2 as

$$\hat{s}_1 = \frac{1}{p} \text{Tr}(\widehat{\Sigma}_0), \quad \hat{s}_2 = \frac{1}{p} \text{Tr}(\widehat{\Sigma}_0)^2.$$

As an important contribution of the paper, we establish the joint asymptotic distribution of the pair $\hat{s}_1$ and $\hat{s}_2$ under the high-dimensional setting.

THEOREM 2.2. *Assume the same conditions as in Theorem 2.1 and assume in addition that*

the limits s_3 and s_4 in (2.2) exist. Then when $\mathbf{x}_t = \Sigma_0^{1/2} \mathbf{z}_t$, we have

$$\begin{pmatrix} \text{Tr}(\widehat{\Sigma}_0) - \text{Tr}(\Sigma_0) \\ \text{Tr}(\widehat{\Sigma}_0^2) - \text{Tr}(\Sigma_0^2) - \frac{1}{T}(\text{Tr}(\Sigma_0))^2 \end{pmatrix} \xrightarrow{d} \mathcal{N}_2 \left(\begin{pmatrix} 0 \\ cs_2(\nu_4 - 2) \end{pmatrix}, \begin{pmatrix} (\nu_4 - 1)cs_2 & (\nu_4 - 1)(2c^2s_1s_2 + 2cs_3) \\ (\nu_4 - 1)(2c^2s_1s_2 + 2cs_3) & 4c^2s_2^2 + (\nu_4 - 1)(4c^3s_1^2s_2 + 8c^2s_1s_3 + 4cs_4) \end{pmatrix} \right).$$

The proof of this theorem is postponed to Section 4. Meanwhile, by the delta method, we easily obtain from the theorem that

$$(2.7) \quad p(\hat{s}_1^2 - s_1^2) \xrightarrow{d} \mathcal{N}(0, 4c(\nu_4 - 1)s_1^2s_2).$$

Now we split the original data $\{\mathbf{x}_t, t = 1, \dots, T\}$ into two halves of length T_1 and T_2 , respectively ($T = T_1 + T_2$). The first half serves to build the test statistic as

$$(2.8) \quad G_{q,1} = \sum_{\tau=1}^q \text{Tr}(\widehat{\Sigma}_{\tau,1} \widehat{\Sigma}_{\tau,1}^*), \quad \widehat{\Sigma}_{\tau,1} = \frac{1}{T_1} \sum_{t=1}^{T_1} \mathbf{x}_t \mathbf{x}_{t-\tau}^* \quad (\mathbf{x}_t = \mathbf{x}_{T_1+t} \text{ for } -T_1 < t \leq 0).$$

With the second half, we build an independent and asymptotically normal estimator $\hat{s}_1^2$ for the characteristic s_1^2 by

$$(2.9) \quad \hat{s}_1 = \frac{1}{p} \text{Tr}(S_{n,2}), \quad \text{where} \quad S_{n,2} = \frac{1}{T_2} \sum_{t=1}^{T_2} \mathbf{x}_{t+T_1} \mathbf{x}_{t+T_1}^*.$$

With Theorem 2.1 applied to $G_{q,1}$, Theorem 2.2 and (2.7) applied to $\hat{s}_1^2$, and the decomposition (2.6), we arrive at the following corollary.

COROLLARY 2.1. *Assume the same conditions as in Theorem 2.2 and that in addition, the dimension p and the sample sizes T_1, T_2 grow to infinity in a related way such that $c_{p,1} = p/T_1 \rightarrow c_1$ and $c_{p,2} = p/T_2 \rightarrow c_2$. Then we have*

$$(2.10) \quad G_{q,1} - qT_1c_{p,1}^2\hat{s}_1^2 \xrightarrow{d} \mathcal{N}(0, \xi^2(c_1, c_2)),$$

where

$$\xi^2(c_1, c_2) = \sigma^2(c_1) + 4q^2c_1^2c_2(\nu_4 - 1)s_1^2s_2 = 2qc_1^2s_2^2 + 4q^2\{2c_1^3 + c_1^2c_2(\nu_4 - 1)\}s_1^2s_2 + 4q^2c_1^3(\nu_4 - 3)s_1^2s_{d,2}.$$

Now we aim at consistent estimates for the unknown quantities in the asymptotic variance $\xi^2(c_1, c_2)$. We observe the following:

- Clearly, $c_{p,1}$ and $c_{p,2}$ are consistent estimates of the limiting ratios c_1 and c_2 , respectively.
- Let

$$\tilde{s}_1 = \frac{1}{p} \text{Tr} \widehat{\Sigma}_0, \quad \text{where} \quad \widehat{\Sigma}_0 = \frac{1}{T} \sum_{t=1}^T \mathbf{x}_t \mathbf{x}_t^*,$$

and

$$\tilde{s}_2 = \frac{1}{p} \text{Tr} \widehat{\Sigma}_0^2 - c_p \tilde{s}_1^2.$$

Then by Theorem 2.2, $\{\tilde{s}_1, \tilde{s}_2\}$ are consistent estimators of $\{s_1, s_2\}$, respectively.

- For non-Gaussian data, $\nu_4 \neq 3$ and we have to estimate the characteristic

$$s_{d,2} = \frac{1}{p} \sum_{i=1}^p d_i^2,$$

where $d_i = \Sigma_{0,ii}$ is the i th diagonal element of Σ_0 . By its very definition, d_i can be consistently estimated by its sample counterpart

$$\tilde{d}_i = \frac{1}{T} \sum_{t=1}^T x_{it}^2.$$

It follows that a consistent estimator for $s_{d,2}$ is simply $\tilde{s}_{d,2} = p^{-1} \sum_{i=1}^p \tilde{d}_i^2$.

In summary, when Σ_0 is unknown, we obtain a procedure for testing the null hypothesis of white noise (2.1) as follows:

$$(2.11) \quad \text{Reject } H_0 \text{ if } \{G_{q,1} - qT_1 c_{p,1}^2 \tilde{s}_1^2 > Z_\alpha \tilde{\xi}\}$$

where

$$\tilde{\xi}^2 = 2qc_{p,1}^2 \tilde{s}_2^2 + 4q^2(2c_{p,1}^3 + c_{p,1}^2 c_{p,2}(\nu_4 - 1)) \tilde{s}_1^2 \tilde{s}_2 + 4q^2 c_{p,1}^3 (\nu_4 - 3) \tilde{s}_1^2 \tilde{s}_{d,2}.$$

Lastly, we recommend the setting of $T_1 = T_2 = [T/2]$ in the absence of any prior information on the unknown characteristics s_1, s_2 and $s_{d,2}$.

2.3. Estimation of the fourth moment ν_4 of a non-Gaussian innovation. To reach a finally usable test in practise, it remains to obtain a consistent estimate for the fourth moment ν_4 of the innovations if they are not Gaussian. This is again a non trivial problem which has not been touched yet in the literature (to our best knowledge). In order to get rid of the role of the unknown cross-sectional covariance matrix Σ_0 , we adopt the splitting strategy of last section: the original data $\{\mathbf{x}_t, t = 1, \dots, T\}$ are split into two halves of length T_1 and T_2 , respectively ($T = T_1 + T_2$). Define the two corresponding sample cross-sectional covariance matrices

$$(2.12) \quad S_{n,1} = \frac{1}{T_1} \sum_{t=1}^{T_1} \mathbf{x}_t \mathbf{x}_t^*, \quad S_{n,2} = \frac{1}{T_2} \sum_{t=1}^{T_2} \mathbf{x}_{t+T_1} \mathbf{x}_{t+T_1}^*.$$

This yields the corresponding F -ratio, or Fisher matrix, $F_n = S_{n,1}^{-1} S_{n,2}$. Observe that this matrix does not depend on the value of the cross-sectional covariance Σ_0 so that in what follows we can assume $\Sigma_0 = I$.

Let $(\lambda_j)_{1 \leq j \leq p}$ be the eigenvalues of F_n . Define K test functions $f_k(x) = \log(a_k + b_k x)$ where $(a_k, b_k)_{1 \leq k \leq K}$ are some positive constants. For each k , we have an eigenvalue statistic of the Fisher matrix

$$X_{T,k} = f_k(\lambda_1) + \cdots + f_k(\lambda_p) - p \int f_k(x) dF_{c_{p,1}, c_{p,2}}(x),$$

where $c_{p,i} = p/T_i$ ($i = 1, 2$) and $F_{c,c'}$ is the limiting Wachter distribution with index (c, c') , see formula (3.1) in (39). It is proved on page 452 of the reference, when p, T_1, T_2 grow proportionally to infinity,

$$(2.13) \quad X_{T,k} = u_{T,k} + v_{T,k}v_4 + \varepsilon_{T,k},$$

where $\{u_{T,k}, v_{T,k}\}$ are computable constants function of $\{c_{p,i}\}$ and (a_k, b_k) , and $\{\varepsilon_{T,k}\}$ is a centered and asymptotically Gaussian error. Then the least squares estimator of v_4 using the above regression model leads to a consistent estimate, say $\hat{v}_4$ for the unknown parameter.

Under the null hypothesis, the observations are independent. We may repeat this estimation procedure, say B times, by taking B random splits of the initial sample. The final estimate of v_4 is then taken to be the average of the estimates $\{\hat{v}_{4,b}\}_{1 \leq b \leq B}$.

Finally we implement in practise the following test procedure for the null hypothesis of white noise (2.1):

$$(2.14) \quad \text{Reject } H_0 \text{ if } \{G_{q,1} - qT_1c_{p,1}^2\hat{s}_1^2 > Z_\alpha\hat{\xi}\}$$

where

$$\hat{\xi}^2 = 2qc_{p,1}^2\tilde{s}_2^2 + 4q^2(2c_{p,1}^3 + c_{p,1}^2c_{p,2}(\hat{v}_4 - 1))\tilde{s}_1^2\tilde{s}_2 + 4q^2c_{p,1}^3(\hat{v}_4 - 3)\tilde{s}_1^2\tilde{s}_{d,2},$$

with the above estimator $\hat{v}_4$ for the fourth moment.

2.4. Tests when the observations are complex-valued. To proceed, we first define $\mathbf{x}_t = \Sigma_0^{1/2} \mathbf{z}_t$ where $\mathbf{z}_t$ is a *proper* complex random vector, and $\Sigma_0^{1/2}$ is such that $\Sigma_0^{1/2}$ is Hermitian with $\Sigma_0 = \Sigma_0^{1/2}(\Sigma_0^{1/2})^*$. *Properness* of a complex random vector $\mathbf{z}_t$ entails that $\mathbb{E}(\mathbf{z}_t \mathbf{z}_t^T) = 0$. We immediately have

$$0 = \mathbb{E}(\mathbf{z}_t \mathbf{z}_t^T) = \mathbb{E}(z_{it}^2)I_p,$$

so that $\mathbb{E}(z_{it}^2) = 0$ for all $i = 1, \dots, p$ and $t = 1, \dots, T$. It also means that $b = |\mathbb{E}(z_{it}^2)|^2 = 0$. Since $\mathbf{x}_t = \Sigma_0^{1/2} \mathbf{z}_t$, we have

$$\mathbb{E}(\mathbf{x}_t \mathbf{x}_t^T) = \mathbb{E}(\Sigma_0^{1/2} \mathbf{z}_t \mathbf{z}_t^T \Sigma_0^{1/2}) = 0,$$

so that we are also assuming an observed vector $\mathbf{x}_t$ is proper.

From Corollary 4.1, since $b = 0$ from the properness of $\mathbf{z}_t$, the asymptotic covariance of G_q is then

$$\text{Var}(G_q) \rightarrow qc^2s_2^2 + 4q^2c^3s_1^2[(v_4 - 2)s_{d,2} - s_2' + 2s_{r,2}],$$

where $s_2' = \lim_{p \rightarrow \infty} \text{Tr}(\Sigma_0 \Sigma_0^T)/p$, $s_{r,2} = \lim_{p \rightarrow \infty} \text{Tr}(\Re^2(\Sigma_0))/p$, with $\Re(A) = (\Re(a_{ij}))$, the matrix of the real parts of all entries in A .

Using Lemma A.1, defining $\mathfrak{I}(A) = (\mathfrak{I}(a_{ij}))$ to be the matrix of the imaginary parts of all entries in A , we have

$$\begin{aligned} 2\text{Tr}(\Re^2(\Sigma_0)) - \text{Tr}(\Sigma_0 \Sigma_0^T) &= 2\text{Tr}(\Sigma_0 \Re(\Sigma_0)) - \text{Tr}(\Sigma_0(\Re(\Sigma_0) - i\mathfrak{I}(\Sigma_0))) \\ &= \text{Tr}(\Sigma_0(\Re(\Sigma_0) + i\mathfrak{I}(\Sigma_0))) = \text{Tr}(\Sigma_0^2), \end{aligned}$$

so that $2s_{r,2} - s'_2 = s_2$. The asymptotic variance for G_q is then

$$\text{Var}(G_q) \rightarrow \sigma^2(c) = qc^2s_2^2 + 4q^2c^3s_1^2[(v_4 - 2)s_{d,2} + s_2],$$

which can be estimated consistently using the estimators suggested in Section 2.2. The test can then be proceeded as described in Section 2.2 using a data splitting scheme, with asymptotic variance $\sigma^2(c_1) + 4q^2c_1^2c_2(v_4 - 1)s_1^2s_2$.

3. Simulation experiments. Most of the experiments of this section are designed in order to compare the test procedures in (2.5) and (2.14) based on the statistics G_q and $G_{q,1}$, with two well known classical white noise tests, namely the Hosking test (17) and the Li-McLeod test (24).

To introduce the Hosking and Li-McLeod tests and using their notations, consider a p -dimensional VARMA(u, v) process of the form

$$\mathbf{x}_t - \Phi_1 \mathbf{x}_{t-1} - \cdots - \Phi_u \mathbf{x}_{t-u} = \mathbf{a}_t - \Theta_1 \mathbf{a}_{t-1} - \cdots - \Theta_v \mathbf{a}_{t-v},$$

where $\mathbf{a}_t$ is a p -dimensional white noise with mean zero and variance Σ . Since $\mathbf{x}_t$ is observed, with an initial guess of u and v , by assuming $\mathbf{a}_t$ to be Gaussian, estimation of parameters $\{\Phi, \Theta\}$ is conducted by the method of maximum likelihood. The initial estimates of u and v are further refined at the diagnostic checking stage based on the autocovariance matrices $\hat{C}_\tau$ of the residuals $\{\hat{\mathbf{a}}_t\}$:

$$\hat{C}_\tau = \frac{1}{T} \sum_{t=\tau+1}^T \hat{\mathbf{a}}_t \hat{\mathbf{a}}_{t-\tau}^*, \quad \tau = 0, 1, 2, \dots$$

Hosking (17) proposed the portmanteau statistic

$$\tilde{Q}_q = T^2 \sum_{\tau=1}^q \frac{1}{T-\tau} \text{Tr}(\hat{C}_\tau^* \hat{C}_0^{-1} \hat{C}_\tau \hat{C}_0^{-1}),$$

while Li et al. (24) recommended the use of the statistic

$$Q_q^* = T \sum_{\tau=1}^q \text{Tr}(\hat{C}_\tau^* \hat{C}_0^{-1} \hat{C}_\tau \hat{C}_0^{-1}) + \frac{p^2 q(q+1)}{2T}.$$

When $\{\mathbf{x}_t\}$ follows a VARMA(u, v) model, both $\tilde{Q}_q$ and Q_q^* converge to $\chi^2(p^2(q - u - v))$ distribution as $T \rightarrow \infty$, while the dimension p remains fixed.

To compare with our multi-lag q test statistics G_q and $G_{q,1}$ when Σ_0 is either known or unknown, we set $u = v = 0$. All tests use 5% significance level and the critical regions of the three tests are as follows:

- (i) G_q when all the limiting parameters are known as defined in (2.5) with $\alpha = 5\%$;
- (ii) $G_{q,1}$ with estimated limiting parameters as defined in (2.14) with $\alpha = 5\%$;
- (iii) Hosking test: $\left\{ \widetilde{Q}_q > \chi_{0.05, qp^2}^2 \right\}$;
- (iv) Li-McLeod test: $\left\{ Q_q^* > \chi_{0.05, qp^2}^2 \right\}$.

Here $Z_{0.05}$ and $\chi_{0.05, m}^2$ denote the upper-5% quantile of the standard normal distribution and the chi-squared distribution with degrees of freedom m , respectively. Empirical statistics are obtained using 2000 independent replicates.

3.1. Empirical sizes. The data is generated as $\mathbf{x}_t = \Sigma_0^{1/2} \mathbf{z}_t$, with $\mathbf{z}_t$, $t = 1, \dots, T$ being independent and identically distributed. We adopt diverse settings for $\mathbf{z}_t$ and Σ_0 respectively to compare the sizes of four test statistics.

As for $\mathbf{z}_t$, we use an indicator *gaussflag* to represent different distributions for $\mathbf{z}_t$:

- (1) *Gaussflag*=0 : $\mathbf{z}_t \sim \mathcal{N}_p(\mathbf{0}, \mathbf{I}_p)$, i.i.d. $t = 1, \dots, T$;
- (2) *Gaussflag*=1 : $\mathbf{z}_t$ with i.i.d components $z_{it} \sim \text{Gamma}(4, 0.5) - 2$, $i = 1, \dots, p$, $t = 1, \dots, T$,
 $\mathbb{E}(z_{it}) = 0$, $\text{Var}(z_{it}) = 1$, $v_4(z_{it}) = 4.5$;
- (3) *Gaussflag*=2 : $\mathbf{z}_t$ with i.i.d components $z_{it} \sim t(6)/\sqrt{1.5}$, $i = 1, \dots, p$, $t = 1, \dots, T$,
 $\mathbb{E}(z_{it}) = 0$, $\text{Var}(z_{it}) = 1$, $v_4(z_{it}) = 6$;

As for Σ_0 , we use an indicator *sigmaflag* to represent different models for Σ_0 :

- (4) *Sigmaflag*=0 : $\Sigma_0 = \mathbf{I}_p$;
- (5) *Sigmaflag*=1 : Σ_0 is diagonal matrix with diagonal entries be an arithmetic sequence running from 0.25 to 6.25.
- (6) *Sigmaflag*=2 : $\Sigma_0 = AA^*$, A is $p \times p$ matrix with entries $a_{ij} \sim U(-1, 1)$ i.i.d..

Table 1 compares the sizes of the four tests for two different q when $\Sigma_0 = \mathbf{I}_p$. Cases when $p > T$ are not considered here since $\widetilde{Q}_q$ and Q_q^* are not applicable then.

The main information from Table 1 is that classical test procedures derived using large sample scheme, namely by letting the sample size $T \rightarrow \infty$ while the dimension p remains fixed, are heavily biased when the dimension p is in fact not negligible with respect to the sample size. To be more precise, these biases are clearly present when the dimension-to-sample ratio p/T is not “small enough”, say greater than 0.1. Such high-dimensional traps for classical procedures have already been reported in other testing problems, see for example (3) and (37). Here we observe that the empirical sizes of the Hosking’s and the Li-McLeod tests quickly degenerate to 0 as the ratio p/T increases from 0.1 to 0.5. In other words, the critical values from their $\chi_{qp^2}^2$ asymptotic limits are seemingly *too large*. On the other hand, the statistics G_q and $G_{q,1}$ have reasonable sizes when compared to the 5% nominal level across all the tested (p, T) combinations. Various (p, T) combinations are accommodated to testify the adaptability of our test statistics, G_q and $G_{q,1}$. Test sizes in both high and low dimension cases are shown in Table 6 and 7. It can be seen that both G_q and $G_{q,1}$ attain the nominal level accurately under various scenarios.

3.2. Empirical powers and adjusted powers. In this section, we compare the empirical powers of the tests by assuming that $\mathbf{x}_t = \Sigma_0^{1/2} \mathbf{y}_t$, $\mathbf{y}_t$ follows a vector autoregressive process of order

1,

$$\mathbf{x}_t = \Sigma_0^{1/2} \mathbf{y}_t, \mathbf{y}_t = A \mathbf{y}_{t-1} + \mathbf{z}_t,$$

where $A = a\mathbf{I}_p$, $\mathbf{z}_t \sim N_p(\mathbf{0}, \mathbf{I}_p)$ being independent of each other for $t = 1, \dots, T$. Here we assign $a = 0.1$ and apply two classic test procedures, $\tilde{Q}_q$ and Q_q^* , to get the power values as in Table 2.

From Table 1 we know that the two classic tests become seriously biased when the dimension p is larger compared to the sample size T . Their sizes approach zero when p/T becomes larger. From Table 2, we see that due to the biased critical values used in $\tilde{Q}_q$ and Q_q^* as shown in Table 1, their powers are driven downward. This is particularly severe when the ratio p/T is larger than 0.5.

In Table 3 and 4, we compare the *intrinsic powers* of all four test procedures when Σ_0 is either spherical or general respectively. Namely, we empirically find the 95 percentiles of $\tilde{Q}_q$ and Q_q^* and use these values as the corrected critical value for the power comparison. It is interesting to observe that after such correction, both $\tilde{Q}_q$ and Q_q^* show very reasonable powers which all increase to 1 when the dimension and the sample size increases. Our test statistics G_q and $G_{q,1}$ also maintain comparably high power in all the tested (p, T) combinations. Table 8 and 9 demonstrates the feasibility of our test statistics under both high and low dimension cases. Comparison with the Hosking's and the Li-McLeod tests sheds new light on the superiority of our test statistics in both low and high dimensional cases. Meanwhile, with true parameters $s_1, s_2, s_{d,2}$, G_q performs reasonably better than $G_{q,1}$ in general. The power loss of $G_{q,1}$ can also be ascribe to sample size reduction.

3.3. Why both the Hosking's and the Li-McLeod tests fail in high dimension?. The experiments here are designed to explore the reasons behind the failure of the Hosking's and the Li-McLeod tests in high dimension. For the test statistics $\tilde{Q}_q$ and Q_q^* as well as our test statistic ϕ_τ , we consider their empirical mean, variance and the 95% quantile, say θ_{emp} , with their theoretical values predicted by their respective asymptotic distributions (denoted as θ_{theo}). As for the two classical tests, we have often observed very large discrepancy between these values so it is more convenient to report the corresponding relative errors $(\theta_{theo} - \theta_{emp})/\theta_{emp}$ (in percentage). This is done in Table 5. It clearly appears from this table that for both statistics $\tilde{Q}_q$ and Q_q^* , the traditional asymptotic theory severely overestimated their variances, that is their empirical means are close to the degree of freedom $p^2(q-u-v)$ of the asymptotic chi-squared distribution while their empirical variances are much smaller than $2p^2(q-u-v)$ as suggested by the same chi-squared limit. This leads to an inflated 95th percentiles which, although in a lesser proportion, is enough to create a high down-bias in the empirical sizes of these two classical tests with high-dimensional data; See Table 1.

3.4. Comparison with other test statistics. In this section, we compare our test statistics with some others in recent literature. Chang, Yao and Zhou (9) proposed an omnibus test for vector white noise using the maximum absolute autocorrelations and cross-correlations of the component series. Let

$$\hat{\Gamma}(k) = \{\hat{\rho}_{ij}(k)\}_{1 \leq i, j \leq p} = \text{diag}\{\hat{\Sigma}(0)\}^{-1/2} \hat{\Sigma}(k) \text{diag}\{\hat{\Sigma}(0)\}^{-1/2}$$

be the sample autocorrelation matrix at lag k , where $\hat{\Sigma}(k) = \frac{1}{T} \sum_{t=1}^{T-k} \mathbf{x}_{t+k} \mathbf{x}_t^*$. Their test statistic T_n is defined as

$$T_n = \max_{1 \leq k \leq q} T_{n,k},$$

where $T_{n,k} = \max_{1 \leq i, j \leq p} T^{1/2} |\hat{\rho}_{ij}(k)|$. Another test statistic T_n^* is defined in the same manner as T_n , only that the time series principal component analysis proposed by Chang *et al.* (8) is applied to the data $\{\mathbf{x}_t\}$ first.

Here we fix $p = 20$, $T = 100$ and adopt the spherical AR(1) process for power comparison, i.e. $\mathbf{x}_t = \Sigma_0^{1/2} \mathbf{y}_t$, $\mathbf{y}_t = A \mathbf{y}_{t-1} + \mathbf{z}_t$, $A = a \mathbf{I}_p$, where $\mathbf{z}_t$ and Σ_0 follow different combinations of settings, i.e. (1)&(2) and (4)&(6). Power values of all the four test statistics, i.e. G_q , $G_{q,1}$, T_n and T_n^* , are compared when VAR coefficient a grows from 0 to 0.5. Empirical statistics are obtained using 2000 independent replicates. Results are shown in Fig 1 and 2.

It can be seen that our test statistics show better performance under this spherical AR(1) model setting. Designed via Frobenius norm of sample autocovariance matrices, the strength of our test statistics are fully demonstrated in such VAR(1) settings. While T_n and T_n^* are more adapted to settings where majority coordinates of the test sequence $\mathbf{x}_t$ or their linear transformations remain to be white noise, see the model settings in Chang, Yao and Zhou (9). Moreover, it can be seen that test size of T_n is a little biased when $p = 20$, $T = 100$. Actually, such bias appears to be more significant when we increase the dimension-to-sample ratio p/T to a relative higher level, say 0.5. On the contrary, our test statistics maintain the nominal level accurately in both low and high dimensional settings. T_n^* shows very resilient powerful performance while it is quite time-consuming due to its relatively complicated bootstrap procedures. All in all, our test statistics G_q and $G_{q,1}$ provide very satisfactory alternatives for high dimensional diagnostic checking.

4. Proofs.

4.1. *Proof of Theorem 2.1.* To derive the null distribution of G_q when $\mathbf{x}_t = \Sigma_0^{1/2} \mathbf{z}_t$, we looked into the Free probability and moment method proposed by Bhattacharjee and Bose (7). In Section 4.2.3 of Bhattacharjee and Bose (7), they have proved the following result:

PROPOSITION 4.1. *Let $\Pi := \Pi(\widehat{\Sigma}_\tau, \widehat{\Sigma}_\tau^* : \tau \geq 0)$ be a symmetric polynomial in $\{\widehat{\Sigma}_\tau, \widehat{\Sigma}_\tau^* : \tau \geq 0\}$,*

$$\sigma_\Pi^2 = \lim \mathbb{E}(\text{Tr}(\Pi) - \mathbb{E}(\text{Tr}(\Pi)))^2.$$

they have proven that,

$$\lim \mathbb{E}(\text{Tr}(\Pi) - \mathbb{E}(\text{Tr}(\Pi)))^k = \begin{cases} 0, & \text{if } k = 2d - 1, \\ \left(\prod_{l=1}^d (2d - 2l + 1) \right) \sigma_\Pi^{2d}, & \text{if } k = 2d. \end{cases}$$

therefore, as $p, T \rightarrow \infty$, $c_p = p/T \rightarrow c \in (0, \infty)$,

$$\text{Tr}(\Pi) - \mathbb{E}\text{Tr}(\Pi) \xrightarrow{d} \mathcal{N}(0, \sigma_\Pi^2).$$

Since G_q is a symmetric polynomial in $\{\widehat{\Sigma}_\tau, \widehat{\Sigma}_\tau^* : \tau \geq 0\}$ and the asymptotic normality of have already been proven, we can directly calculate the first two order moments of the test statistic to get the null distribution. This is done in the following corollary which is a direct consequence of moment calculations presented in the appendix, Section A.

COROLLARY 4.1. *Let the assumptions for $\mathbf{z}_i$ in Theorem 2.1 hold. Under the framework $p/T \rightarrow c > 0$, assume that $\|\Sigma_0\| = O(1)$. Then as $p, T \rightarrow \infty$,*

$$\mathbb{E}(G_q) \sim qp^2 s_1^2 / T,$$

$$\text{Var}(G_q) \rightarrow qc^2(s_2^2 + b^2(s_2')^2) + 4q^2c^3(\nu_4 - b - 2)s_1^2s_{d,2} + 8q^2c^3s_1^2s_{r,2} + 4q^2c^3(b - 1)s_1^2s_2',$$

where $s_2' = \lim_{p \rightarrow \infty} \text{Tr}(\Sigma_0 \Sigma_0^T) / p$, $s_{r,2} = \lim_{p \rightarrow \infty} \text{Tr}(\Re^2(\Sigma_0)) / p$.

If the z_{it} 's are real, then Σ_0 is real symmetric and $b = 1$, $s_2' = s_{r,2} = s_2$. The asymptotic formula for $\text{Var}(G_q)$ then reduces to

$$2qc^2s_2^2 + 4q^2c^3(\nu_4 - 3)s_1^2s_{d,2} + 8q^2c^3s_1^2s_2,$$

which further reduces to $2qc^2s_2^2 + 8q^2c^3s_1^2s_2$ if all the z_{it} 's are Gaussian.

4.2. *Proof of Theorem 2.2.* For any $n \times n$ Hermitian matrix M with real eigenvalues $\lambda_1, \dots, \lambda_n$, the empirical spectral distribution (ESD for short) of M is defined by $F^M = n^{-1} \sum_{j=1}^n \delta_{\lambda_j}$, where δ_a denotes the Dirac mass at a . The Stieltjes transform of any distribution G is defined as

$$m_G(z) = \int \frac{1}{x - z} dG(x), \quad \Im(z) > 0,$$

where $\Im(z)$ stands for the imaginary part of z .

Bai and Silverstein (2) and Zheng et al. (40) studied the central limit theorem for linear spectral statistics of large-dimensional sample covariance matrices. By their notation, let $B_n = (1/N)T_n^{1/2}X_nX_n^*T_n^{1/2}$, where $X_n = (X_{ij})$ is $n \times N$ with i.i.d. complex standardized entries having finite fourth moment, $T_n^{1/2}$ is a Hermitian square root of the nonnegative definite Hermitian matrix T_n . It has been proven that if for all n, i, j , $X_{i,j}^n$ are independent, with probability 1, $F^{T_n} \xrightarrow{d} H$, a proper cumulative distribution function(c.d.f.) and $n/N \rightarrow y > 0$ as $n \rightarrow \infty$, then with probability 1, ESD of B_n , F^{B_n} converges in distribution to $F^{y,H}$, a non-random proper c.d.f. If $\underline{B}_n = (1/N)X_n^*T_nX_n$, then its LSD $\underline{F}^{y,H}$ satisfies

$$\underline{F}^{y,H} = (1 - y)I_{[0,\infty)} + yF^{y,H},$$

and its Stieltjes transform has inverse

$$(4.1) \quad z = -\frac{1}{\underline{m}} + y \int \frac{t}{1 + \underline{m}t} dH(t).$$

Define

$$G_n(x) = n \left[F^{B_n}(x) - F^{y_n, H_n}(x) \right],$$

and $f_1, \dots, f_k$ be functions on $\mathbb{R}$ analytic, assume $\mathbb{E}(X_{ij}) = 0$, $\mathbb{E}(|X_{ij}|^2) = 1$, $\mathbb{E}(|X_{ij}|^4) = \nu_4 < \infty$, then random vector

$$(4.2) \quad \left(\int f_1(x) dG_n(x), \dots, \int f_k(x) dG_n(x) \right),$$

forms a tight sequence in n and (4.2) converges weakly to a Gaussian vector $(X_{f_1}, \dots, X_{f_k})$ with means

(4.3)

$$\mathbb{E}X_f = -\frac{1}{2\pi i} \oint f(z) \frac{y \int \frac{\underline{m}(z)^3 t^2}{(1+t\underline{m}(z))^3} dH(t)}{\left(1 - y \int \frac{\underline{m}(z)^2 t^2}{(1+t\underline{m}(z))^2} dH(t)\right)^2} dz - \frac{\nu_4 - 3}{2\pi i} \oint f(z) \frac{y \int \frac{\underline{m}(z)^3 t^2}{(1+t\underline{m}(z))^3} dH(t)}{1 - y \int \frac{\underline{m}(z)^2 t^2}{(1+t\underline{m}(z))^2} dH(t)} dz,$$

and covariance function

(4.4)

$$\begin{aligned} \text{Cov}(X_f, X_g) = & -\frac{1}{2\pi^2} \iint \frac{f(z_1)g(z_2)}{(\underline{m}(z_1) - \underline{m}(z_2))^2} \frac{d}{dz_1} \underline{m}(z_1) \frac{d}{dz_2} \underline{m}(z_2) dz_1 dz_2 \\ & - \frac{y(\nu_4 - 3)}{4\pi^2} \iint f(z_1)g(z_2) \left(\int \frac{t}{(t\underline{m}(z_1) + 1)^2} \cdot \frac{t}{(t\underline{m}(z_2) + 1)^2} dH(t) \right) d\underline{m}(z_1) d\underline{m}(z_2), \end{aligned}$$

($f, g \in \{f_1, \dots, f_k\}$). The contours in (4.3) and (4.4) (two in (4.4), which we may assume to be non-overlapping) are closed and are taken in the positive direction in the complex plane, each enclosing the support of $F^{y,H}$.

Our target is to derive the joint distribution of $(\text{Tr}(\widehat{\Sigma}_0), \text{Tr}(\widehat{\Sigma}_0^2))$, where $\widehat{\Sigma}_0 = \frac{1}{T} \sum_{t=1}^T \mathbf{x}_t \mathbf{x}_t^*$, $\mathbf{x}_t = \Sigma_0^{1/2} \mathbf{z}_t$, $\mathbf{z}_t$ has independent components z_{it} satisfying $\mathbb{E}z_{it} = 0$, $\mathbb{E}z_{it}^2 = 1$, $\mathbb{E}z_{it}^4 = \nu_4 < \infty$. We can see that $(\text{Tr}(\widehat{\Sigma}_0), \text{Tr}(\widehat{\Sigma}_0^2))$ are actually two linear spectral statistics of the sample covariance matrix $\widehat{\Sigma}_0$. Let $T_n = \Sigma_0$, $X_n = (\mathbf{z}_1, \dots, \mathbf{z}_T)$, $p \leftrightarrow n$, $T \leftrightarrow N$, $(c = p/T) \leftrightarrow (y = n/N)$, then

$$\begin{aligned} B_n &= \frac{1}{N} T_n^{1/2} X_n X_n^* T_n^{1/2} = \frac{1}{T} \sum_{t=1}^T \Sigma_0^{1/2} \mathbf{z}_t \mathbf{z}_t^* \Sigma_0^{1/2} = \frac{1}{T} \sum_{t=1}^T \mathbf{x}_t \mathbf{x}_t^* = \widehat{\Sigma}_0, \\ f_1(x) &= x, \quad f_2(x) = x^2, \quad \left(\int f_1(x) dG_n(x), \int f_2(x) dG_n(x) \right) = (\text{Tr}(\widehat{\Sigma}_0), \text{Tr}(\widehat{\Sigma}_0^2)), \end{aligned}$$

thus what's left to do is to derive the corresponding centering term, mean term (4.3) and variance term (4.4) for the two functions $f_1(x)$ and $f_2(x)$ when $B_n = \widehat{\Sigma}_0$.

To start with, we determine the centering terms

$$n \int f_j(x) dF^{y_n, H_n}(x) = p \int x^j dF^{c_n, H_n}(x), \quad j = 1, 2.$$

There are the first two moments of the distribution F^{c_n, H_n} . These moments are simple functions of the first two moments of the distribution of H_n , see e.g. (38), as follows:

$$\begin{aligned} \int x dF^{c_n, H_n}(x) &= \int x dH_n(x) = \frac{1}{p} \text{Tr}(\Sigma_0) \\ \int x^2 dF^{c_n, H_n}(x) &= \int x^2 dH_n(x) + c_n \left\{ \int x dH_n(x) \right\}^2 = \frac{1}{p} \text{Tr}(\Sigma_0^2) + c_n \left\{ \frac{1}{p} \text{Tr}(\Sigma_0) \right\}^2. \end{aligned}$$

Multiplying these moments by p yield the required centering terms.

Now we consider the limiting parameters in (4.3) and (4.4). The conclusion follows from the following calculation of limiting parameters:

$$(4.5) \quad \mathbb{E}X_{f_1} = 0, \quad \text{Var}(X_{f_1}) = c(\nu_4 - 1)s_2,$$

$$(4.6) \quad \mathbb{E}X_{f_2} = c(\nu_4 - 2)s_2, \quad \text{Var}(X_{f_2}) = 4c^2s_2^2 + (\nu_4 - 1)(4c^3s_1^2s_2 + 8c^2s_1s_3 + 4cs_4),$$

$$(4.7) \quad \text{Cov}(X_{f_1}, X_{f_2}) = (\nu_4 - 1)(2c^2s_1s_2 + 2cs_3).$$

The details of these calculations are given in the appendix, Section B.

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TABLE 1
Empirical sizes for our tests G_q and $G_{q,1}$, Hosking test $\tilde{Q}_q$ and Li-McLeod test Q_q^ .*

p	T	p/T	G_q		$G_{q,1}$		$\tilde{Q}_q$		Q_q^*	
			$q = 1$	$q = 3$	$q = 1$	$q = 3$	$q = 1$	$q = 3$	$q = 1$	$q = 3$
5	1000	0.005	0.0640	0.0645	0.0650	0.0685	0.0490	0.0478	0.0488	0.0476
10	2000	0.005	0.0625	0.0525	0.0650	0.0560	0.0492	0.0440	0.0492	0.0436
25	5000	0.005	0.0465	0.0545	0.0590	0.0545	0.0498	0.0528	0.0498	0.0528
40	8000	0.005	0.0490	0.0475	0.0430	0.0515	0.0508	0.0520	0.0508	0.0520
10	1000	0.01	0.0630	0.0455	0.0620	0.0565	0.0472	0.0468	0.0470	0.0464
20	2000	0.01	0.0585	0.0555	0.0525	0.0530	0.0502	0.0530	0.0502	0.0530
50	5000	0.01	0.0505	0.0500	0.0490	0.0490	0.0488	0.0498	0.0488	0.0498
80	8000	0.01	0.0565	0.0415	0.0475	0.0505	0.0464	0.0406	0.0464	0.0404
50	1000	0.05	0.0565	0.0590	0.0555	0.0590	0.0408	0.0466	0.0408	0.0466
100	2000	0.05	0.0435	0.0420	0.0500	0.0515	0.0432	0.0414	0.0432	0.0414
250	5000	0.05	0.0445	0.0610	0.0510	0.0490	0.0456	0.0436	0.0456	0.0434
400	8000	0.05	0.0500	0.0455	0.0475	0.0495	0.0418	0.0410	0.0418	0.0410
10	100	0.1	0.0605	0.0590	0.0655	0.0715	0.0300	0.0400	0.0280	0.0362
40	400	0.1	0.0495	0.0525	0.0485	0.0500	0.0362	0.0342	0.0358	0.0338
60	600	0.1	0.0525	0.0550	0.0490	0.0515	0.0340	0.0340	0.0340	0.0338
100	1000	0.1	0.047	0.0515	0.0605	0.0575	0.0370	0.0268	0.0366	0.0264
50	100	0.5	0.0630	0.0540	0.0565	0.0585	0.0006	0.0018	0.0006	0.0018
200	400	0.5	0.0530	0.0520	0.0560	0.0565	0.0010	0.0004	0.0010	0.0004
300	600	0.5	0.0495	0.0480	0.0525	0.0525	0.0002	0.0008	0.0002	0.0008
500	1000	0.5	0.0475	0.0455	0.0580	0.0600	0	0	0	0
90	100	0.9	0.0560	0.0605	0.0480	0.0490	0	0	0	0
180	200	0.9	0.0455	0.0455	0.0535	0.0525	0	0	0	0
540	600	0.9	0.0500	0.0480	0.0580	0.0610	0	0	0	0
900	1000	0.9	0.0490	0.0550	0.0465	0.0505	0	0	0	0

TABLE 2
Power performance of two classical test statistics: Hosking test $\tilde{Q}_q$ and Li-McLeod test Q_q^ .*

p	T	p/T	$\tilde{Q}_q$		Q_q^*	
			$q = 1$	$q = 3$	$q = 1$	$q = 3$
10	100	0.1	0.0952	0.0952	0.0914	0.0926
20	200	0.1	0.2392	0.1994	0.2362	0.1958
40	400	0.1	0.6638	0.5410	0.6622	0.5380
60	600	0.1	0.9406	0.8452	0.9404	0.8448
100	1000	0.1	1	0.9982	1	0.9982
50	100	0.5	0.0014	0.0060	0.0014	0.0052
100	200	0.5	0.0036	0.0208	0.0030	0.0194
200	400	0.5	0.0330	0.2022	0.0328	0.1994
300	600	0.5	0.1156	0.6348	0.1138	0.6306
500	1000	0.5	0.5816	0.9974	0.5806	0.9972
80	100	0.8	0	0	0	0
160	200	0.8	0	0	0	0
320	400	0.8	0	0	0	0
480	600	0.8	0	0	0	0
800	1000	0.8	0.0004	0.0038	0.0004	0.0032
90	100	0.9	0	0	0	0
180	200	0.9	0	0	0	0
360	400	0.9	0	0	0	0
540	600	0.9	0	0	0	0
900	1000	0.9	0	0	0	0

TABLE 3
Adjusted powers of Hosking test $\tilde{Q}_q$ and Li-McLeod test Q_q^* as compared to powers of our tests G_q and $G_{q,1}$ when $\Sigma_0 = \mathbf{I}_p$

p	T	p/T	a	G_q		$G_{q,1}$		$\tilde{Q}_q$		Q_q^*	
				$q = 1$	$q = 3$	$q = 1$	$q = 3$	$q = 1$	$q = 3$	$q = 1$	$q = 3$
5	1000	0.005	0.05	0.4755	0.2810	0.2455	0.1380	0.4290	0.2535	0.4290	0.2540
10	2000	0.005	0.05	0.9190	0.6310	0.5015	0.2610	0.9010	0.6495	0.9010	0.6500
25	5000	0.005	0.05	1	0.9995	0.9910	0.7580	1	0.9985	1	0.9985
40	8000	0.005	0.05	1	1	1	0.9825	1	1	1	1
10	1000	0.01	0.05	0.5025	0.2790	0.2310	0.1340	0.4715	0.2450	0.4715	0.2450
20	2000	0.01	0.05	0.9315	0.6340	0.5010	0.2410	0.9315	0.6530	0.9315	0.6530
50	5000	0.01	0.05	1	0.9975	0.9930	0.7535	1	0.9990	1	0.9990
80	8000	0.01	0.05	1	1	1	0.9740	1	1	1	1
50	1000	0.05	0.05	0.5475	0.2760	0.2050	0.1110	0.4795	0.2910	0.4795	0.2905
100	2000	0.05	0.05	0.9680	0.6750	0.4880	0.2225	0.9470	0.7450	0.9470	0.7460
250	5000	0.05	0.05	1	1	0.9905	0.7165	1	1	1	1
400	8000	0.05	0.05	1	1	1	0.9715	1	1	1	1
10	100	0.1	0.1	0.1980	0.1370	0.1290	0.1090	0.1392	0.1218	0.1392	0.1214
40	400	0.1	0.1	0.8295	0.5325	0.3235	0.1915	0.7082	0.5968	0.7082	0.5988
60	600	0.1	0.1	0.9910	0.8045	0.5875	0.3270	0.9598	0.8808	0.9598	0.8816
100	1000	0.1	0.1	1	0.9970	0.9130	0.5875	1	0.9992	1	0.9992
200	2000	0.1	0.1	1	1	1	0.9730	1	1	1	1
50	100	0.5	0.1	0.2210	0.1610	0.1115	0.0985	0.1004	0.1376	0.1004	0.1380
200	400	0.5	0.1	0.9350	0.7795	0.3925	0.2900	0.4012	0.7708	0.4012	0.7708
300	600	0.5	0.1	0.9990	0.9755	0.6660	0.5320	0.6626	0.9746	0.6626	0.9748
500	1000	0.5	0.1	1	1	0.9675	0.8740	0.9666	1	0.9666	1
1000	2000	0.5	0.1	1	1	1	1	1	1	1	1
90	100	0.9	0.1	0.2465	0.2025	0.1115	0.1000	0.1384	0.0992	0.1384	0.1002
360	400	0.9	0.1	0.9630	0.8950	0.5015	0.4150	0.7138	0.5172	0.7138	0.5176
540	600	0.9	0.1	1	0.9980	0.7885	0.6830	0.9496	0.8368	0.9496	0.8368
900	1000	0.9	0.1	1	1	0.9915	0.9685	0.9998	0.9966	0.9998	0.9966
1800	2000	0.9	0.1	1	1	1	1	1	1	1	1

TABLE 4
Adjusted powers of Hosking test $\tilde{Q}_q$ and Li-McLeod test Q_q^* as compared to powers of our tests G_q and $G_{q,1}$ when $\Sigma_0 = AA^*$, $A = (a_{ij})$, $a_{ij} \sim U(-1, 1)$ i.i.d.

p	T	p/T	a	G_q		$G_{q,1}$		$\tilde{Q}_q$		Q_q^*	
				$q = 1$	$q = 3$	$q = 1$	$q = 3$	$q = 1$	$q = 3$	$q = 1$	$q = 3$
5	1000	0.005	0.05	0.4605	0.2885	0.2500	0.1610	0.4895	0.2715	0.4895	0.2715
10	2000	0.005	0.05	0.8420	0.5855	0.4845	0.2900	0.9030	0.6475	0.9030	0.6475
25	5000	0.005	0.05	1	0.9965	0.9650	0.7340	1	1	1	1
40	8000	0.005	0.05	1	1	0.9995	0.9655	1	1	1	1
10	1000	0.01	0.05	0.4825	0.3050	0.2580	0.1580	0.4595	0.2850	0.4595	0.2845
20	2000	0.01	0.05	0.8825	0.5835	0.4925	0.2740	0.9275	0.6270	0.9275	0.6275
50	5000	0.01	0.05	1	0.9970	0.9780	0.7430	1	0.9990	1	0.9990
80	8000	0.01	0.05	1	1	1	0.9785	1	1	1	1
50	1000	0.05	0.05	0.5285	0.2605	0.2220	0.1355	0.4705	0.3100	0.4705	0.3100
100	2000	0.05	0.05	0.9475	0.6370	0.4990	0.2470	0.9325	0.7235	0.9325	0.7235
250	5000	0.05	0.05	1	1	0.9890	0.7170	1	1	1	1
400	8000	0.05	0.05	1	1	1	0.9705	1	1	1	1
10	100	0.1	0.1	0.2080	0.1475	0.1475	0.1200	0.1510	0.1200	0.1510	0.1210
40	400	0.1	0.1	0.8080	0.4920	0.3760	0.1975	0.7460	0.6095	0.7460	0.6095
60	600	0.1	0.1	0.9805	0.7630	0.5635	0.2975	0.9650	0.8940	0.9650	0.8940
100	1000	0.1	0.1	1	0.9860	0.8935	0.5340	1	0.9985	1	0.9985
200	2000	0.1	0.1	1	1	1	0.9575	1	1	1	1
50	100	0.5	0.1	0.2225	0.1560	0.1270	0.1035	0.1045	0.1420	0.1045	0.1420
200	400	0.5	0.1	0.8850	0.6370	0.3570	0.2370	0.3935	0.7580	0.3935	0.7590
300	600	0.5	0.1	0.9930	0.9190	0.5755	0.3900	0.6965	0.9645	0.6965	0.9645
500	1000	0.5	0.1	1	1	0.9195	0.7295	0.9685	1	0.9685	1
1000	2000	0.5	0.1	1	1	1	0.9990	1	1	1	1
90	100	0.9	0.1	0.2035	0.1600	0.1160	0.1060	0.1105	0.1000	0.1105	0.1005
360	400	0.9	0.1	0.9245	0.7485	0.3990	0.2950	0.7005	0.5100	0.7005	0.5105
540	600	0.9	0.1	0.9990	0.9685	0.6640	0.4985	0.9400	0.8125	0.9400	0.8130
900	1000	0.9	0.1	1	1	0.9540	0.8515	1	0.9990	1	0.9990
1800	2000	0.9	0.1	1	1	1	1	1	1	1	1

TABLE 5

Relative errors for the mean, variance and 95 percentile for Hosking test statistic $\tilde{Q}_q$ and Li-McLeod test statistic Q_q^* (with $q = 3$)

p	T	p/T	$\tilde{Q}_q$			Q_q^*		
			Mean	Variance	95%Quantile	Mean	Variance	95%Quantile
10	100	0.1	0.234%	19.976%	1.366%	0.234%	24.922%	1.547%
20	200	0.1	0.067%	30.862%	0.993%	0.067%	33.526%	1.049%
40	400	0.1	-0.015%	22.057%	0.253%	-0.015%	23.286%	0.265%
60	600	0.1	0.000%	21.457%	0.162%	0.000%	22.269%	0.162%
100	1000	0.1	0.007%	20.666%	0.125%	0.007%	21.153%	0.125%
50	100	0.5	0.041%	267.179%	1.322%	0.041%	282.546%	1.354%
100	200	0.5	0.007%	284.025%	0.655%	0.007%	291.875%	0.662%
200	400	0.5	0.000%	289.080%	0.330%	0.000%	292.998%	0.330%
300	600	0.5	0.000%	297.059%	0.222%	0.000%	299.734%	0.222%
500	1000	0.5	0.000%	296.364%	0.134%	0.000%	297.941%	0.134%
80	100	0.8	0.010%	1742.257%	1.289%	0.005%	1820.096%	1.300%
160	200	0.8	0.000%	2020.024%	0.655%	0.000%	2063.959%	0.657%
320	400	0.8	0.000%	2214.386%	0.332%	0.000%	2237.811%	0.332%
480	600	0.8	0.001%	2266.151%	0.223%	0.001%	2282.093%	0.223%
800	1000	0.8	0.000%	2348.823%	0.137%	0.000%	2358.701%	0.137%
90	100	0.9	0.004%	5382.234%	1.292%	0.000%	5618.993%	1.297%
180	200	0.9	0.000%	6906.920%	0.657%	0.000%	7053.897%	0.658%
360	400	0.9	0.000%	8110.500%	0.332%	0.000%	8195.108%	0.332%
540	600	0.9	0.000%	8705.234%	0.222%	0.000%	8764.569%	0.222%
900	1000	0.9	0.000%	9170.563%	0.133%	0.000%	9208.205%	0.133%

			Gaussflag = 0				Gaussflag = 1				Gaussflag = 2			
p	T	$Sigmaflag$	G_q		$G_{q,1}$		G_q		$G_{q,1}$		G_q		$G_{q,1}$	
			$q = 1$	$q = 3$	$q = 1$	$q = 3$	$q = 1$	$q = 3$	$q = 1$	$q = 3$	$q = 1$	$q = 3$		
50	100	0	0.0490	0.0535	0.0540	0.0580	0.0515	0.0530	0.0580	0.0500	0.0700	0.0595	0.0560	0.0540
100	200	0	0.0595	0.0640	0.0585	0.0540	0.0545	0.0575	0.0530	0.0600	0.0560	0.0510	0.0490	0.0510
200	400	0	0.0430	0.0475	0.0495	0.0475	0.0500	0.0505	0.0565	0.0615	0.0540	0.0480	0.0480	0.0485
100	100	0	0.0545	0.0555	0.0435	0.0450	0.0530	0.0560	0.0580	0.0550	0.0455	0.0430	0.0480	0.0425
200	200	0	0.0555	0.0540	0.0550	0.0560	0.0500	0.0470	0.0580	0.0635	0.0550	0.0485	0.0555	0.0505
400	400	0	0.0435	0.0455	0.0515	0.0510	0.0465	0.0490	0.0570	0.0595	0.0525	0.0565	0.0450	0.0470
200	100	0	0.0435	0.0430	0.0590	0.0585	0.0550	0.0535	0.0540	0.0560	0.0630	0.0605	0.0505	0.0530
400	200	0	0.0470	0.0525	0.0500	0.0485	0.0485	0.0535	0.0515	0.0515	0.0540	0.0480	0.0435	0.0425
800	400	0	0.0480	0.0505	0.0390	0.0415	0.0445	0.0480	0.0480	0.0495	0.0435	0.0450	0.0490	0.0495
50	100	1	0.0545	0.0570	0.0540	0.0540	0.0575	0.0530	0.0725	0.0725	0.0695	0.0695	0.0670	0.0635
100	200	1	0.0490	0.0525	0.0610	0.0540	0.0505	0.0540	0.0685	0.0630	0.0545	0.0540	0.0585	0.0610
200	400	1	0.0425	0.0485	0.0455	0.0500	0.0590	0.0475	0.0605	0.0590	0.0525	0.0555	0.0565	0.0550
100	100	1	0.0500	0.0490	0.0485	0.0535	0.0585	0.0635	0.0585	0.0550	0.0555	0.0515	0.0615	0.0655
200	200	1	0.0540	0.0530	0.0425	0.0495	0.0605	0.0595	0.0475	0.0445	0.0555	0.0525	0.0515	0.0570
400	400	1	0.0585	0.0555	0.0440	0.0460	0.0640	0.0570	0.0490	0.0445	0.0585	0.0545	0.0585	0.0605
200	100	1	0.0530	0.0525	0.0585	0.0570	0.0520	0.0525	0.0555	0.0550	0.0585	0.0545	0.0535	0.0545
400	200	1	0.0555	0.0575	0.0480	0.0475	0.0445	0.0445	0.0510	0.0530	0.0480	0.0470	0.0540	0.0535
800	400	1	0.0505	0.0520	0.0500	0.0520	0.0540	0.0545	0.0490	0.0470	0.0500	0.0495	0.0440	0.0435
50	100	2	0.0525	0.0530	0.0710	0.0670	0.0605	0.0630	0.0510	0.0490	0.0675	0.0600	0.0535	0.0555
100	200	2	0.0510	0.0525	0.0585	0.0550	0.0670	0.0615	0.0525	0.0520	0.0500	0.0555	0.0330	0.0295
200	400	2	0.0440	0.0485	0.0595	0.0570	0.0540	0.0560	0.0410	0.0400	0.0605	0.0590	0.0495	0.0495
100	100	2	0.0565	0.0525	0.0540	0.0605	0.0545	0.0545	0.0465	0.0425	0.0530	0.0600	0.0405	0.0435
200	200	2	0.0555	0.0580	0.0615	0.0585	0.0550	0.0495	0.0440	0.0415	0.0510	0.0525	0.0395	0.0395
400	400	2	0.0530	0.0565	0.0460	0.0465	0.0535	0.0505	0.0405	0.0425	0.0505	0.0540	0.0415	0.0405
200	100	2	0.0555	0.0515	0.0560	0.0570	0.0540	0.0505	0.0400	0.0440	0.0545	0.0535	0.0425	0.0410
400	200	2	0.0575	0.0555	0.0450	0.0415	0.0475	0.0460	0.0510	0.0530	0.0540	0.0535	0.0410	0.0380
800	400	2	0.0435	0.0495	0.0510	0.0500	0.0470	0.0545	0.0470	0.0460	0.0545	0.0550	0.0450	0.0410

TABLE 6
High dimensional test sizes of our tests G_q and $G_{q,1}$

p	T	Sigmaflag	$\text{Gaussflag} = 0$						$\text{Gaussflag} = 1$						$\text{Gaussflag} = 2$					
			G_q		$G_{q,1}$		G_q		$G_{q,1}$		G_q		$G_{q,1}$		G_q		$G_{q,1}$		G_q	
			$q = 1$	$q = 3$	$q = 1$	$q = 3$	$q = 1$	$q = 3$	$q = 1$	$q = 3$	$q = 1$	$q = 3$	$q = 1$	$q = 3$	$q = 1$	$q = 3$	$q = 1$	$q = 3$	$q = 1$	$q = 3$
5	500	0	0.0665	0.0660	0.0700	0.0690	0.0670	0.0615	0.0790	0.0645	0.0600	0.0655	0.0695	0.0700	0.0665	0.0655	0.0695	0.0700	0.0665	0.0655
10	1000	0	0.0560	0.0610	0.0560	0.0595	0.0495	0.0490	0.0590	0.0575	0.0650	0.0620	0.0585	0.0655	0.0650	0.0620	0.0585	0.0655	0.0650	0.0620
20	2000	0	0.0535	0.0560	0.0580	0.0580	0.0540	0.0585	0.0585	0.0525	0.0595	0.0550	0.0595	0.0550	0.0595	0.0550	0.0595	0.0550	0.0595	0.0550
25	500	0	0.0470	0.0445	0.0620	0.0580	0.0365	0.0495	0.0590	0.0565	0.0675	0.0650	0.0555	0.0545	0.0675	0.0650	0.0555	0.0545	0.0675	0.0650
50	1000	0	0.0465	0.0490	0.0625	0.0565	0.0435	0.0470	0.0445	0.0495	0.0545	0.0550	0.0530	0.0530	0.0545	0.0550	0.0530	0.0530	0.0545	0.0550
100	2000	0	0.0615	0.0505	0.0600	0.0630	0.0445	0.0430	0.0420	0.0410	0.0435	0.0535	0.0560	0.0620	0.0435	0.0535	0.0560	0.0620	0.0435	0.0535
50	500	0	0.0530	0.0580	0.0675	0.0680	0.0415	0.0475	0.0595	0.0535	0.0545	0.0565	0.0535	0.0510	0.0545	0.0565	0.0535	0.0510	0.0545	0.0565
100	1000	0	0.0475	0.0495	0.0560	0.0555	0.0615	0.0575	0.0550	0.0540	0.0500	0.0455	0.0515	0.0480	0.0500	0.0455	0.0515	0.0480	0.0500	0.0455
200	2000	0	0.0560	0.0510	0.0580	0.0570	0.0635	0.0615	0.0480	0.0545	0.0590	0.0560	0.0570	0.0565	0.0590	0.0560	0.0570	0.0565	0.0590	0.0560
5	500	1	0.0695	0.0675	0.0745	0.0730	0.0855	0.0685	0.0800	0.0790	0.0690	0.0605	0.0845	0.0700	0.0695	0.0605	0.0845	0.0700	0.0695	0.0605
10	1000	1	0.0670	0.0605	0.0630	0.0635	0.0745	0.0550	0.0665	0.0615	0.0640	0.0585	0.0735	0.0760	0.0670	0.0585	0.0735	0.0760	0.0670	0.0585
20	2000	1	0.0660	0.0635	0.0685	0.0585	0.0535	0.0575	0.0575	0.0570	0.0570	0.0475	0.0560	0.0535	0.0570	0.0475	0.0560	0.0535	0.0570	0.0475
25	500	1	0.0505	0.0495	0.0445	0.0535	0.0650	0.0600	0.0690	0.0715	0.0550	0.0515	0.0615	0.0550	0.0550	0.0515	0.0615	0.0550	0.0550	0.0515
50	1000	1	0.0550	0.0555	0.0505	0.0555	0.0575	0.0535	0.0550	0.0470	0.0650	0.0595	0.0575	0.0535	0.0650	0.0595	0.0575	0.0535	0.0650	0.0595
100	2000	1	0.0605	0.0535	0.0515	0.0595	0.0540	0.0480	0.0540	0.0515	0.0530	0.0540	0.0540	0.0595	0.0530	0.0540	0.0540	0.0595	0.0530	0.0540
50	500	1	0.0580	0.0530	0.0595	0.0670	0.0510	0.0545	0.0650	0.0650	0.0595	0.0550	0.0530	0.0510	0.0595	0.0550	0.0530	0.0510	0.0595	0.0550
100	1000	1	0.0515	0.0460	0.0515	0.0475	0.0670	0.0520	0.0570	0.0520	0.0510	0.0570	0.0575	0.0605	0.0510	0.0570	0.0575	0.0605	0.0510	0.0570
200	2000	1	0.0500	0.0400	0.0550	0.0575	0.0510	0.0535	0.0445	0.0465	0.0555	0.0475	0.0490	0.0455	0.0555	0.0475	0.0490	0.0455	0.0555	0.0475
5	500	2	0.0665	0.0640	0.0670	0.0740	0.0710	0.0690	0.0650	0.0690	0.0585	0.0715	0.0775	0.0705	0.0665	0.0715	0.0775	0.0705	0.0665	0.0715
10	1000	2	0.0535	0.0585	0.0655	0.0685	0.0720	0.0540	0.0660	0.0670	0.0675	0.0615	0.0675	0.0535	0.0535	0.0615	0.0675	0.0535	0.0535	0.0615
20	2000	2	0.0590	0.0580	0.0610	0.0570	0.0605	0.0545	0.0595	0.0475	0.0650	0.0670	0.0680	0.0670	0.0590	0.0670	0.0680	0.0670	0.0590	0.0670
25	500	2	0.0625	0.0630	0.0535	0.0595	0.0695	0.0605	0.0625	0.0595	0.0630	0.0585	0.0600	0.0525	0.0625	0.0585	0.0600	0.0525	0.0625	0.0585
50	1000	2	0.0520	0.0565	0.0500	0.0505	0.0525	0.0550	0.0560	0.0490	0.0560	0.0555	0.0530	0.0450	0.0520	0.0555	0.0530	0.0450	0.0520	0.0555
100	2000	2	0.0605	0.0555	0.0480	0.0610	0.0430	0.0525	0.0465	0.0520	0.0505	0.0510	0.0495	0.0390	0.0505	0.0510	0.0495	0.0390	0.0505	0.0510
50	500	2	0.0515	0.0465	0.0605	0.0585	0.0560	0.0505	0.0645	0.0590	0.0580	0.0530	0.0490	0.0495	0.0515	0.0530	0.0490	0.0495	0.0515	0.0530
100	1000	2	0.0480	0.0515	0.0530	0.0505	0.0550	0.0500	0.0495	0.0415	0.0575	0.0530	0.0400	0.0395	0.0480	0.0530	0.0400	0.0395	0.0480	0.0530
200	2000	2	0.0495	0.0515	0.0555	0.0525	0.0445	0.0490	0.0395	0.0390	0.0505	0.0550	0.0495	0.0425	0.0495	0.0550	0.0495	0.0425	0.0495	0.0550

TABLE 7

Low dimensional test sizes of our tests G_q and $G_{q,1}$

$a = 0.15$			$Gauss\ flag = 0$				$Gauss\ flag = 1$				$Gauss\ flag = 2$			
p	T	$Sigma\ flag$	G_q		$G_{q,1}$		G_q		$G_{q,1}$		G_q		$G_{q,1}$	
			$q = 1$	$q = 3$	$q = 1$	$q = 3$	$q = 1$	$q = 3$	$q = 1$	$q = 3$	$q = 1$	$q = 3$		
50	100	0	0.5455	0.3960	0.2150	0.1780	0.4385	0.3050	0.1505	0.1250	0.3780	0.2435	0.1510	0.1260
100	200	0	0.9560	0.8355	0.4580	0.3575	0.8955	0.7000	0.3330	0.2460	0.8010	0.5635	0.3040	0.2270
200	400	0	1	1	0.9155	0.7870	1	0.9945	0.7620	0.6015	1	0.9700	0.6620	0.4970
100	100	0	0.6485	0.5310	0.2355	0.2055	0.5060	0.3980	0.1835	0.1550	0.4130	0.3095	0.1530	0.1325
200	200	0	0.9875	0.9550	0.5645	0.4840	0.9440	0.8520	0.3900	0.3325	0.8555	0.7270	0.3155	0.2735
400	400	0	1	1	0.9745	0.9490	1	1	0.8670	0.7880	1	0.9985	0.7675	0.6655
200	100	0	0.8075	0.7415	0.3035	0.2795	0.6420	0.5550	0.2125	0.1960	0.5320	0.4565	0.1735	0.1630
400	200	0	1	0.9985	0.7235	0.6765	0.9885	0.9695	0.5200	0.4710	0.9530	0.9015	0.4045	0.3630
800	400	0	1	1	0.9975	0.9955	1	1	0.9720	0.9590	1	1	0.9120	0.8695
50	100	1	0.5240	0.3630	0.2040	0.1605	0.4175	0.2705	0.1670	0.1370	0.3365	0.2095	0.1505	0.1245
100	200	1	0.9225	0.7530	0.4300	0.3165	0.8265	0.5930	0.3040	0.2135	0.7700	0.5070	0.2640	0.1955
200	400	1	1	0.9980	0.8675	0.6900	1	0.9815	0.7200	0.5270	0.9985	0.9420	0.6095	0.4355
100	100	1	0.5700	0.4500	0.2165	0.1780	0.4585	0.3405	0.1595	0.1300	0.3980	0.2695	0.1435	0.1200
200	200	1	0.9725	0.9060	0.4755	0.3890	0.9185	0.7595	0.3775	0.3005	0.8075	0.6180	0.2650	0.2120
400	400	1	1	1	0.9370	0.8675	1	0.9990	0.8005	0.6755	1	0.9895	0.6795	0.5400
200	100	1	0.6980	0.6085	0.2600	0.2380	0.5620	0.4580	0.1940	0.1740	0.4755	0.3895	0.1515	0.1345
400	200	1	0.9945	0.9825	0.6460	0.5770	0.9660	0.9155	0.4605	0.3950	0.9145	0.8180	0.3625	0.3160
800	400	1	1	1	0.9905	0.9790	1	1	0.9270	0.8780	1	1	0.8235	0.7480
50	100	2	0.4945	0.3335	0.2190	0.1720	0.4660	0.3100	0.1750	0.1270	0.4220	0.2715	0.1315	0.1035
100	200	2	0.8990	0.7090	0.4180	0.2895	0.8800	0.6245	0.3420	0.2390	0.8520	0.5855	0.2635	0.1785
200	400	2	1	0.9970	0.8460	0.6335	1	0.9845	0.7330	0.4990	1	0.9660	0.6585	0.4170
100	100	2	0.5705	0.4245	0.2110	0.1765	0.4920	0.3315	0.1700	0.1310	0.4285	0.2940	0.1440	0.1150
200	200	2	0.9555	0.8300	0.4545	0.3595	0.9180	0.7575	0.3490	0.2625	0.8835	0.6825	0.3045	0.2260
400	400	2	1	1	0.9140	0.8000	1	0.9980	0.8170	0.6620	1	0.9910	0.7095	0.5320
200	100	2	0.6575	0.5315	0.2410	0.2135	0.5655	0.4470	0.1790	0.1580	0.4905	0.3845	0.1525	0.1245
400	200	2	0.9890	0.9635	0.5640	0.4905	0.9645	0.9010	0.4485	0.3775	0.9355	0.8375	0.3330	0.2730
800	400	2	1	1	0.9725	0.9400	1	1	0.9155	0.8365	1	1	0.8485	0.7655

TABLE 8
High dimensional test power of our tests G_q and $G_{q,1}$

$\alpha = 0.1$			$Gaussflag = 0$						$Gaussflag = 1$						$Gaussflag = 2$					
p	T	$\Sigma flag$	G_q		$G_{q,1}$		G_q		$G_{q,1}$		G_q		$G_{q,1}$		G_q		$G_{q,1}$		G_q	
			$q = 1$	$q = 3$	$q = 1$	$q = 3$	$q = 1$	$q = 3$	$q = 1$	$q = 3$	$q = 1$	$q = 3$	$q = 1$	$q = 3$	$q = 1$	$q = 3$	$q = 1$	$q = 3$	$q = 1$	$q = 3$
5	500	0	0.8730	0.6125	0.4775	0.2830	0.8365	0.5580	0.4415	0.2375	0.8410	0.5390	0.4285	0.2245	0.8410	0.5390	0.4285	0.2245	0.8410	0.5390
10	1000	0	1	0.9745	0.9115	0.5675	1	0.9535	0.8585	0.5070	0.9990	0.9525	0.8285	0.4585	0.9990	0.9525	0.8285	0.4585	0.9990	0.9525
20	2000	0	1	1	1	0.9605	1	1	0.9985	0.9270	1	1	0.9985	0.8985	1	1	0.9985	0.8985	1	1
25	500	0	0.9380	0.6470	0.4830	0.2555	0.9145	0.5795	0.3905	0.2140	0.8970	0.5085	0.3335	0.1630	0.8970	0.5085	0.3335	0.1630	0.8970	0.5085
50	1000	0	1	0.9910	0.9180	0.5530	1	0.9630	0.8075	0.4235	1	0.9470	0.7645	0.3400	1	0.9470	0.7645	0.3400	1	0.9470
100	2000	0	1	1	1	0.9640	1	1	0.9995	0.8810	1	1	0.9990	0.7950	1	1	0.9990	0.7950	1	1
50	500	0	0.9525	0.6890	0.4375	0.2440	0.9180	0.5610	0.3495	0.1885	0.8875	0.5060	0.2985	0.1675	0.8875	0.5060	0.2985	0.1675	0.8875	0.5060
100	1000	0	1	0.9955	0.9030	0.5920	1	0.9730	0.8030	0.4525	1	0.9370	0.7100	0.3680	1	0.9370	0.7100	0.3680	1	0.9370
200	2000	0	1	1	1	0.9755	1	1	0.9990	0.8940	1	1	0.9925	0.7865	1	1	0.9925	0.7865	1	1
5	500	1	0.7770	0.5775	0.4455	0.2805	0.7565	0.5320	0.4280	0.2695	0.7775	0.5460	0.4420	0.2675	0.7775	0.5460	0.4420	0.2675	0.7775	0.5460
10	1000	1	0.9960	0.9375	0.8180	0.5515	0.9960	0.9305	0.8185	0.5170	0.9955	0.9245	0.7845	0.4740	0.9955	0.9245	0.7845	0.4740	0.9955	0.9245
20	2000	1	1	1	0.9990	0.9425	1	1	0.9975	0.9120	1	1	0.9995	0.8810	1	1	0.9995	0.8810	1	1
25	500	1	0.9190	0.6350	0.4695	0.2510	0.8855	0.5635	0.4220	0.2190	0.8595	0.4920	0.3785	0.1970	0.8595	0.4920	0.3785	0.1970	0.8595	0.4920
50	1000	1	1	0.9815	0.8905	0.5400	1	0.9595	0.8365	0.4280	1	0.9435	0.8010	0.3685	1	0.9435	0.8010	0.3685	1	0.9435
100	2000	1	1	1	1	0.9565	1	1	1	0.8820	1	1	0.9985	0.7810	1	1	0.9985	0.7810	1	1
50	500	1	0.9405	0.6565	0.4420	0.2450	0.9065	0.5690	0.3860	0.1905	0.8740	0.4870	0.3225	0.1680	0.8740	0.4870	0.3225	0.1680	0.8740	0.4870
100	1000	1	1	0.9900	0.8840	0.5270	1	0.9755	0.8160	0.4340	1	0.9240	0.7310	0.3280	1	0.9240	0.7310	0.3280	1	0.9240
200	2000	1	1	1	1	0.9555	1	1	1	0.8805	1	1	0.9970	0.7905	1	1	0.9970	0.7905	1	1
5	500	2	0.7650	0.5265	0.4260	0.2750	0.7900	0.5450	0.4345	0.2645	0.7595	0.5345	0.4220	0.2630	0.7595	0.5345	0.4220	0.2630	0.7595	0.5345
10	1000	2	0.9905	0.9180	0.7985	0.5275	0.9895	0.9090	0.7910	0.5225	0.9895	0.9115	0.7790	0.4890	0.9895	0.9115	0.7790	0.4890	0.9895	0.9115
20	2000	2	1	1	0.9980	0.9340	1	1	0.9985	0.9130	1	1	0.9945	0.9085	1	1	0.9945	0.9085	1	1
25	500	2	0.8780	0.6095	0.4710	0.2495	0.8760	0.5895	0.4500	0.2140	0.8810	0.5815	0.4290	0.1950	0.8810	0.5815	0.4290	0.1950	0.8810	0.5815
50	1000	2	1	0.9795	0.8905	0.5340	1	0.9685	0.8735	0.4610	1	0.9610	0.8195	0.4210	1	0.9610	0.8195	0.4210	1	0.9610
100	2000	2	1	1	1	0.9515	1	1	1	0.8960	1	1	1	0.8790	1	1	1	0.8790	1	1
50	500	2	0.9210	0.6450	0.4705	0.2660	0.9125	0.6240	0.4090	0.2045	0.8990	0.5715	0.3615	0.1655	0.8990	0.5715	0.3615	0.1655	0.8990	0.5715
100	1000	2	1	0.9890	0.9035	0.5595	1	0.9755	0.8585	0.4530	1	0.9635	0.7990	0.3715	1	0.9635	0.7990	0.3715	1	0.9635
200	2000	2	1	1	1	0.9620	1	1	0.9995	0.9215	1	1	0.9975	0.8475	1	1	0.9975	0.8475	1	1

TABLE 9

Low dimensional test power of our tests G_q and $G_{q,1}$

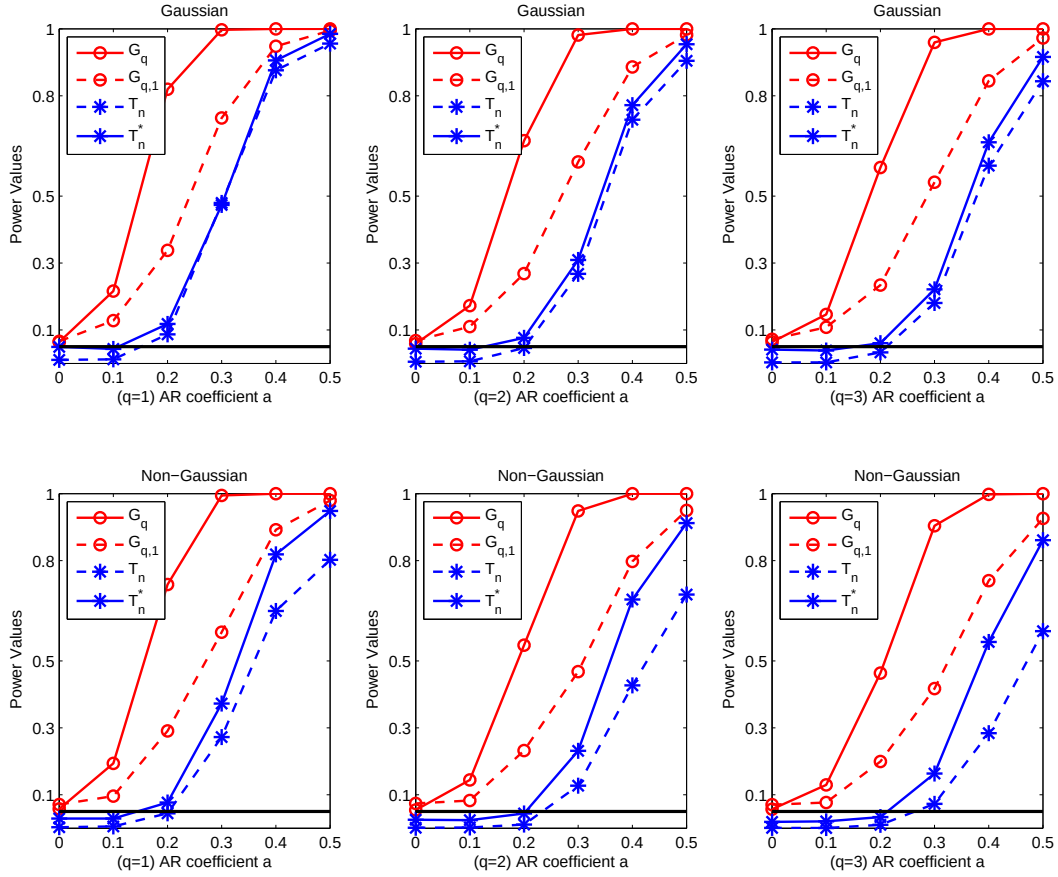


Fig 1: $p = 20$, $T = 100$, Sigmaflag=0, i.e. $\Sigma_0 = I_p$, 2000 replicates

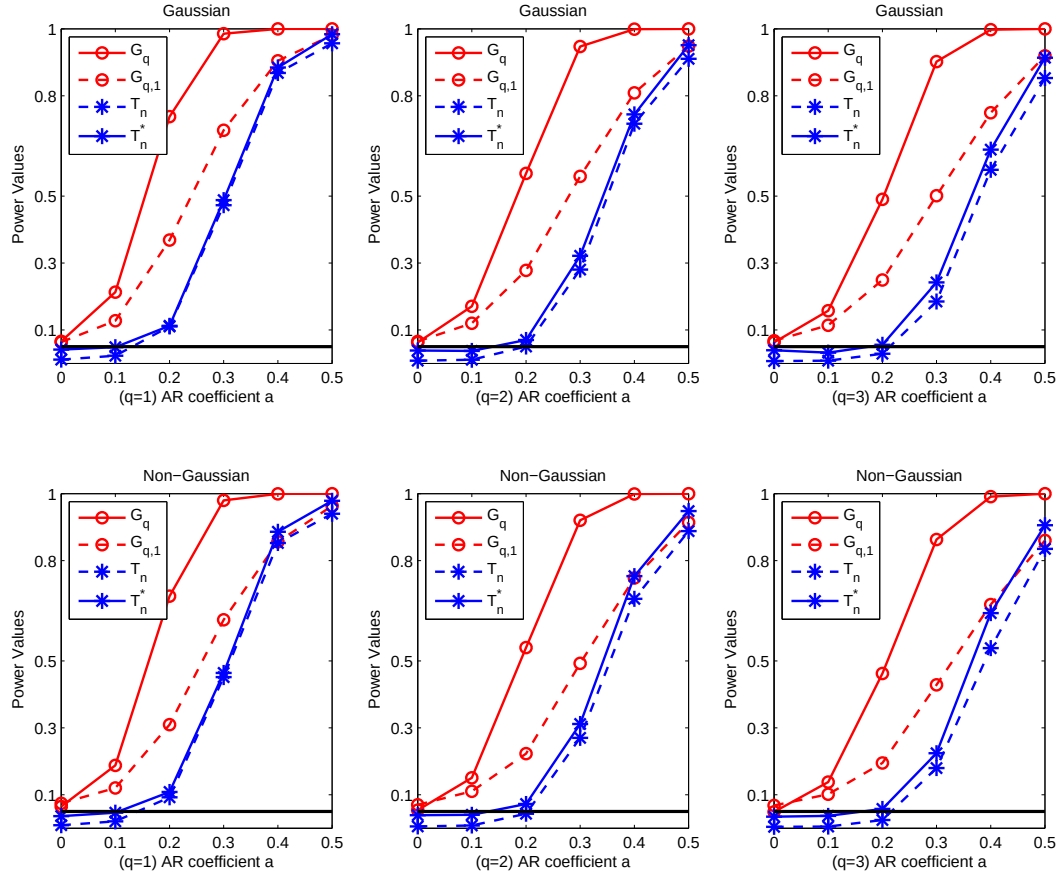


Fig 2: $p = 20$, $T = 100$, Sigmaflag=2, i.e. $\Sigma_0 = AA^*$, $A = (a_{ij})$, $a_{ij} \sim U(-1, 1)$, 2000 replicates

Appendixes

APPENDIX A: TECHNICAL LEMMAS

Hereafter, for a matrix $A = (a_{ij})$, define $\Re(A) = (\Re(a_{ij}))$ and $\Im(A) = (\Im(a_{ij}))$, where $\Re(z)$ and $\Im(z)$ are the real and imaginary parts of the complex number z .

LEMMA A.1. *Let A be Hermitian. Then $\text{Tr}(A\Re(A)) = \text{Tr}(\Re^2(A))$.*

PROOF. Let a_{ij} be the (i, j) th element of A . Then

$$\text{Tr}(A\Re(A)) = \sum_{j,k} a_{jk} \Re(a_{kj}) = \sum_{j,k} \Re^2(a_{kj}) - i \sum_{j,k} \Re(a_{kj}) \Im(a_{kj}) = \text{Tr}(\Re^2(A)),$$

where we used $\Re(a_{jk}) = \Re(a_{kj})$ and $\Im(a_{jk}) = -\Im(a_{kj})$. $\square$

LEMMA A.2. *Let A be Hermitian and $\mathbf{z}_t$ has independent components with identical second and fourth order moments. Furthermore, $\mathbb{E}(z_{it}) = 0$, $\mathbb{E}|z_{it}|^2 = 1$, $\mathbb{E}|z_{it}|^4 = \nu_4 < \infty$ and $b = |\mathbb{E}(z_{it}^2)|^2$. Then*

$$\mathbb{E}(\mathbf{z}_t \mathbf{z}_t^* A \mathbf{z}_t \mathbf{z}_t^*) = \text{Tr}(A) I_p + (\nu_4 - b - 2) \text{diag}(A) + 2\Re(A) + (b - 1)A^T.$$

If A is symmetric, then by denoting the k th moment of z_{it} as $\mathbb{E}(z^k)$, $k = 2, 4$,

$$\mathbb{E}(\mathbf{z}_t \mathbf{z}_t^T A \mathbf{z}_t \mathbf{z}_t^T) = \mathbb{E}^2(z^2) \text{Tr}(A) I_p + (\mathbb{E}(z^4) - 3\mathbb{E}^2(z^2)) \text{diag}(A) + 2\mathbb{E}^2(z^2)A.$$

PROOF. The (i, j) th entry of $\mathbb{E}(\mathbf{z}_t \mathbf{z}_t^* A \mathbf{z}_t \mathbf{z}_t^*)$ is

$$\begin{aligned} (\mathbb{E}(\mathbf{z}_t \mathbf{z}_t^* A \mathbf{z}_t \mathbf{z}_t^*))_{ij} &= \mathbb{E}\left(\sum_{k,\ell} z_{it} z_{kt}^* a_{k\ell} z_{\ell t} z_{jt}^*\right) = \sum_k a_{kk} \mathbb{E}(z_{it} |z_{kt}|^2 z_{jt}^*) + \sum_{k \neq \ell} a_{k\ell} \mathbb{E}(z_{it} z_{kt}^* z_{\ell t} z_{jt}^*) \\ &= \begin{cases} a_{ij} + a_{ji} \mathbb{E}(z_{it}^2) \mathbb{E}((z_{jt}^*)^2), & i \neq j; \\ (\nu_4 - 1)a_{ii} + \text{Tr}(A), & i = j \end{cases} = \begin{cases} 2\Re(a_{ij}) + a_{ji}(|\mathbb{E}(z_{it}^2)|^2 - 1), & i \neq j; \\ (\nu_4 - 1)a_{ii} + \text{Tr}(A), & i = j. \end{cases} \end{aligned}$$

This completes the proof of the first part. For the second part,

$$\begin{aligned} (\mathbb{E}(\mathbf{z}_t \mathbf{z}_t^T A \mathbf{z}_t \mathbf{z}_t^T))_{ij} &= \mathbb{E}\left(\sum_{k,\ell} z_{it} z_{kt} a_{k\ell} z_{\ell t} z_{jt}\right) = \sum_k a_{kk} \mathbb{E}(z_{kt}^2 z_{it} z_{jt}) + \sum_{k \neq \ell} a_{k\ell} \mathbb{E}(z_{it} z_{kt} z_{\ell t} z_{jt}) \\ &= \begin{cases} (a_{ij} + a_{ji}) \mathbb{E}^2(z^2), & i \neq j; \\ (\mathbb{E}(z^4) - \mathbb{E}^2(z^2))a_{ii} + \mathbb{E}^2(z^2) \text{Tr}(A), & i = j \end{cases} \\ &= \begin{cases} 2a_{ij} \mathbb{E}^2(z^2), & i \neq j; \\ (\mathbb{E}(z^4) - \mathbb{E}^2(z^2))a_{ii} + \mathbb{E}^2(z^2) \text{Tr}(A), & i = j \end{cases} \end{aligned}$$

This completes the proof of the lemma. $\square$

LEMMA A.3. *Let the assumptions for $\mathbf{z}_t$ in Lemma A.2 hold. Then for $s \neq t$ and $\tau = 1, \dots, q$,*

$$\begin{aligned}
V_1 &= \mathbb{E}(\mathbf{z}_t^* \Sigma_0 \mathbf{z}_t) = \text{Tr}(\Sigma_0), \\
V_2 &= \mathbb{E}(\mathbf{z}_t^* \Sigma_0 \mathbf{z}_t)^2 = \text{Tr}^2(\Sigma_0) + (\nu_4 - b - 2)\text{Tr}(\text{diag}^2(\Sigma_0)) + 2\text{Tr}(\Re^2(\Sigma_0)) + (b - 1)\text{Tr}(\Sigma_0 \Sigma_0^T), \\
V_3 &= \mathbb{E}(\mathbf{z}_t^* \Sigma_0 \mathbf{z}_s)^2 = b\text{Tr}(\Sigma_0 \Sigma_0^T), \\
V'_3 &= \mathbb{E}|\mathbf{z}_t^* \Sigma_0 \mathbf{z}_s|^2 = \text{Tr}(\Sigma_0^2), \\
V_4 &= \mathbb{E}((\mathbf{z}_t^* \Sigma_0 \mathbf{z}_{t+\tau})^2 (\mathbf{z}_t^* \Sigma_0 \mathbf{z}_{t-\tau})^2) \\
&= b^2 \text{Tr}^2(\Sigma_0 \Sigma_0^T) + (\mathbb{E}(\bar{z}^4) \mathbb{E}^2(z^2) - 3b^2) \text{Tr}(\text{diag}^2(\Sigma_0 \Sigma_0^T)) + 2b^2 \text{Tr}(\Sigma_0 \Sigma_0^T)^2, \\
V'_4 &= \mathbb{E}(|\mathbf{z}_{t+\tau}^* \Sigma_0 \mathbf{z}_t|^2 |\mathbf{z}_t^* \Sigma_0 \mathbf{z}_{t-\tau}|^2) \\
&= \text{Tr}^2(\Sigma_0^2) + (\nu_4 - b - 2)\text{Tr}(\text{diag}^2(\Sigma_0^2)) + 2\text{Tr}(\Re^2(\Sigma_0^2)) + (b - 1)\text{Tr}(\Sigma_0^2 (\Sigma_0^2)^T).
\end{aligned}$$

PROOF. We have

$$V_1 = \text{Tr}[\mathbb{E}(\mathbf{z}_t \mathbf{z}_t^*) \Sigma_0] = \text{Tr}(\Sigma_0).$$

For V_2 , using Lemma A.1 and the first part of Lemma A.2,

$$\begin{aligned}
V_2 &= \text{Tr}[\mathbb{E}(\mathbf{z}_t \mathbf{z}_t^* \Sigma_0 \mathbf{z}_t \mathbf{z}_t^*) \Sigma_0] \\
&= \text{Tr} \left(\text{Tr}(\Sigma_0) \Sigma_0 + (\nu_4 - b - 2) \text{diag}(\Sigma_0) \Sigma_0 + 2\Re(\Sigma_0) \Sigma_0 + (b - 1) \Sigma_0^T \Sigma_0 \right) \\
&= \text{Tr}^2(\Sigma_0) + (\nu_4 - b - 2) \text{Tr}(\text{diag}^2(\Sigma_0)) + 2\text{Tr}(\Re^2(\Sigma_0)) + (b - 1) \text{Tr}(\Sigma_0 \Sigma_0^T).
\end{aligned}$$

Also,

$$V_3 = \mathbb{E}(\mathbf{z}_t^* \Sigma_0 \mathbf{z}_s \mathbf{z}_s^T \Sigma_0^T \bar{\mathbf{z}}_t) = \text{Tr}(\mathbb{E}(\bar{\mathbf{z}}_t \mathbf{z}_t^*) \Sigma_0 \mathbb{E}(\mathbf{z}_s \mathbf{z}_s^T) \Sigma_0^T) = b \text{Tr}(\Sigma_0 \Sigma_0^T).$$

For V'_3 ,

$$V'_3 = \mathbb{E}(\mathbf{z}_t^* \Sigma_0 \mathbf{z}_s \mathbf{z}_s^* \Sigma_0 \mathbf{z}_t) = \text{Tr}(\mathbb{E}(\mathbf{z}_t \mathbf{z}_t^*) \Sigma_0 \mathbb{E}(\mathbf{z}_s \mathbf{z}_s^*) \Sigma_0) = \text{Tr}(\Sigma_0^2).$$

Using the second part of Lemma A.2 with $A = \Sigma_0 \mathbf{z}_{t+\tau} \mathbf{z}_{t+\tau}^T \Sigma_0^T$ and z replaced by $\bar{z}$,

$$\begin{aligned}
V_4 &= \mathbb{E}(\mathbf{z}_t^* \Sigma_0 \mathbf{z}_{t+\tau} \mathbf{z}_{t+\tau}^T \Sigma_0^T \bar{\mathbf{z}}_t \mathbf{z}_t^* \Sigma_0 \mathbf{z}_{t-\tau} \mathbf{z}_{t-\tau}^T \Sigma_0^T \bar{\mathbf{z}}_t) = \text{Tr}(\mathbb{E}(\bar{\mathbf{z}}_t \bar{\mathbf{z}}_t^T \Sigma_0 \mathbf{z}_{t+\tau} \mathbf{z}_{t+\tau}^T \Sigma_0^T \bar{\mathbf{z}}_t \bar{\mathbf{z}}_t^T) \Sigma_0 \mathbb{E}(\mathbf{z}_{t-\tau} \mathbf{z}_{t-\tau}^T) \Sigma_0^T) \\
&= \mathbb{E}(z^2) \text{Tr} \left(\mathbb{E}^2(\bar{z}^2) \text{Tr}(\mathbb{E}(\Sigma_0 \mathbf{z}_{t+\tau} \mathbf{z}_{t+\tau}^T \Sigma_0^T)) \Sigma_0 \Sigma_0^T + (\mathbb{E}(\bar{z}^4) - 3\mathbb{E}^2(\bar{z}^2)) \text{diag}(\mathbb{E}(\Sigma_0 \mathbf{z}_{t+\tau} \mathbf{z}_{t+\tau}^T \Sigma_0^T)) \Sigma_0 \Sigma_0^T \right. \\
&\quad \left. + 2\mathbb{E}^2(\bar{z}^2) \mathbb{E}(\Sigma_0 \mathbf{z}_{t+\tau} \mathbf{z}_{t+\tau}^T \Sigma_0^T) \Sigma_0 \Sigma_0^T \right) \\
&= b^2 \text{Tr}^2(\Sigma_0 \Sigma_0^T) + (\mathbb{E}(\bar{z}^4) \mathbb{E}^2(z^2) - 3b^2) \text{Tr}(\text{diag}^2(\Sigma_0 \Sigma_0^T)) + 2b^2 \text{Tr}(\Sigma_0 \Sigma_0^T)^2.
\end{aligned}$$

Finally, using the first part of Lemma A.2 with $A = \Sigma_0 \mathbf{z}_{t+\tau} \mathbf{z}_{t+\tau}^* \Sigma_0$,

$$\begin{aligned}
V'_4 &= \mathbb{E}(\mathbf{z}_t^* \Sigma_0 \mathbf{z}_{t+\tau} \mathbf{z}_{t+\tau}^* \Sigma_0 \mathbf{z}_t \mathbf{z}_t^* \Sigma_0 \mathbf{z}_{t-\tau} \mathbf{z}_{t-\tau}^* \Sigma_0 \mathbf{z}_t) = \text{Tr}(\mathbb{E}(\mathbf{z}_t \mathbf{z}_t^* \Sigma_0 \mathbf{z}_{t+\tau} \mathbf{z}_{t+\tau}^* \Sigma_0 \mathbf{z}_t \mathbf{z}_t^*) \Sigma_0 \mathbb{E}(\mathbf{z}_{t-\tau} \mathbf{z}_{t-\tau}^*) \Sigma_0) \\
&= \text{Tr} \left(\text{Tr}(\mathbb{E}(\Sigma_0 \mathbf{z}_{t+\tau} \mathbf{z}_{t+\tau}^* \Sigma_0)) \Sigma_0^2 + (\nu_4 - b - 2) \text{diag}(\mathbb{E}(\Sigma_0 \mathbf{z}_{t+\tau} \mathbf{z}_{t+\tau}^* \Sigma_0)) \Sigma_0^2 \right. \\
&\quad \left. + 2\Re(\mathbb{E}(\Sigma_0 \mathbf{z}_{t+\tau} \mathbf{z}_{t+\tau}^* \Sigma_0)) \Sigma_0^2 + (b - 1) (\mathbb{E}(\Sigma_0 \mathbf{z}_{t+\tau} \mathbf{z}_{t+\tau}^* \Sigma_0))^T \Sigma_0^2 \right) \\
&= \text{Tr}^2(\Sigma_0^2) + (\nu_4 - b - 2) \text{Tr}(\text{diag}^2(\Sigma_0^2)) + 2\text{Tr}(\Re^2(\Sigma_0^2)) + (b - 1) \text{Tr}(\Sigma_0^2 (\Sigma_0^2)^T).
\end{aligned}$$

The proof is now complete. $\square$

LEMMA A.4. *Let the assumptions for $\mathbf{z}_t$ in Lemma A.2 hold. Then $G_q = \sum_{\tau=1}^q \text{Tr}(\widehat{\Sigma}_\tau \widehat{\Sigma}_\tau^*)$ has expectation and variance given by*

$$\begin{aligned} \mathbb{E}(G_q) &= qV_1^2/T, \\ \text{Var}(G_q) &= \frac{q(V_2 - V_1^2)^2 + 2q(\Re(V_4) + V_4') + q(T-2)(V_3^2 + (V_3')^2)}{T^3} + \frac{4q^2V_1^2(V_2 - V_1^2)}{T^3}. \end{aligned}$$

PROOF. Write

$$G_q = \sum_{\tau=1}^q Q_\tau, \quad \text{with} \quad Q_\tau = \frac{1}{T^2} \sum_{t,s=1}^T \mathbb{E}(\mathbf{x}_s^* \mathbf{x}_t \mathbf{x}_{t-\tau}^* \mathbf{x}_{s-\tau}).$$

Since $\mathbf{x}_t = \Sigma_0^{1/2} \mathbf{z}_t$ with the $\mathbf{z}_t$'s independent of each other, we have

$$\mathbb{E}(Q_\tau) = \frac{1}{T} \mathbb{E}(\mathbf{x}_t^* \mathbf{x}_t \mathbf{x}_{t-\tau}^* \mathbf{x}_{t-\tau}) = \frac{1}{T} \mathbb{E}^2(\mathbf{x}_t^* \mathbf{x}_t) = \frac{\text{Tr}^2(\Sigma_0)}{T} = V_1^2/T.$$

The value of $\mathbb{E}(G_q)$ follows.

To evaluate $\mathbb{E}(Q_\tau^2)$, observe that

$$\mathbb{E}(Q_\tau^2) = \frac{1}{T^4} \sum_{t_1, t_2, s_1, s_2=1}^T \mathbb{E}(\mathbf{z}_{s_1}^* \Sigma_0 \mathbf{z}_{t_1} \mathbf{z}_{t_1-\tau}^* \Sigma_0 \mathbf{z}_{s_1-\tau} \mathbf{z}_{s_2}^* \Sigma_0 \mathbf{z}_{t_2} \mathbf{z}_{t_2-\tau}^* \Sigma_0 \mathbf{z}_{s_2-\tau}).$$

Let $E(t_1, s_1, t_2, s_2)$ be the expectation on the right hand side above. We detail the cases where this expectation is non-zero below.

(I) $s_1 = t_1, s_2 = t_2$. Sub-cases:

i. $t_1 = t_2 (= t)$:

$$T^{-4} \sum_{t=1}^T E(t, t, t, t) = \frac{1}{T^4} \sum_{t=1}^T \mathbb{E}(\mathbf{z}_t^* \Sigma_0 \mathbf{z}_t)^2 \mathbb{E}(\mathbf{z}_{t-\tau}^* \Sigma_0 \mathbf{z}_{t-\tau})^2 = \frac{1}{T^4} \sum_{t=1}^T V_2^2 = V_2^2/T^3;$$

ii. $t_1 = t_2 - \tau (= t)$:

$$T^{-4} \sum_{t=1}^T E(t, t, t + \tau, t + \tau) = \frac{1}{T^4} \sum_{t=1}^T \mathbb{E}(\mathbf{z}_t^* \Sigma_0 \mathbf{z}_t)^2 \mathbb{E}(\mathbf{z}_{t+\tau}^* \Sigma_0 \mathbf{z}_{t+\tau}) \mathbb{E}(\mathbf{z}_{t-\tau}^* \Sigma_0 \mathbf{z}_{t-\tau}) = V_2 V_1^2/T^3.$$

iii. $t_1 = t_2 + \tau (= t)$:

$$T^{-4} \sum_{t=1}^T E(t, t, t - \tau, t - \tau) = V_2 V_1^2/T^3.$$

iv. Otherwise:

$$\begin{aligned} T^{-4} \sum_{s \neq t, t+\tau, t-\tau} E(t, t, s, s) &= \frac{1}{T^4} \sum_{s \neq t, t+\tau, t-\tau} \mathbb{E}(\mathbf{z}_t^* \Sigma_0 \mathbf{z}_t) \mathbb{E}(\mathbf{z}_{t-\tau}^* \Sigma_0 \mathbf{z}_{t-\tau}) \mathbb{E}(\mathbf{z}_s^* \Sigma_0 \mathbf{z}_s) \mathbb{E}(\mathbf{z}_{s-\tau}^* \Sigma_0 \mathbf{z}_{s-\tau}) \\ &= (T^2 - 3T) V_1^4/T^4. \end{aligned}$$

(II) $s_1 = s_2, t_1 = t_2$. Sub-cases (not overlapping with (I))

i. $s_1 = t_1 + \tau (= t)$:

$$T^{-4} \sum_{t=1}^T E(t - \tau, t, t - \tau, t) = \frac{1}{T^4} \sum_{t=1}^T \mathbb{E}[(\mathbf{z}_{t-2\tau}^* \Sigma_0 \mathbf{z}_{t-2\tau})^2 (\mathbf{z}_t^* \Sigma_0 \mathbf{z}_{t-\tau})^2] = \bar{V}_4/T^3.$$

ii. $s_1 = t_1 - \tau (= t)$:

$$T^{-4} \sum_{t=1}^T E(t + \tau, t, t + \tau, t) = V_4/T^3.$$

iii. Otherwise:

$$T^{-4} \sum_{s \neq t + \tau, t - \tau} E(t, s, t, s) = (T^2 - 2T)V_3^2/T^4.$$

(III) $s_1 = t_2, s_2 = t_1$. Sub-cases (not overlapping with (I) and (II))

i. $s_2 = t_2 + \tau (= t)$:

$$T^{-4} \sum_{t=1}^T E(t, t - \tau, t - \tau, t) = V_4'/T^3.$$

ii. $s_2 = t_2 - \tau (= t)$:

$$T^{-4} \sum_{t=1}^T E(t, t + \tau, t + \tau, t) = V_4'/T^3.$$

iii. Otherwise:

$$T^{-4} \sum_{s \neq t + \tau, t - \tau} E(t, s, s, t) = (T^2 - 2T)(V_3')^2/T^4.$$

With the above,

$$\begin{aligned} \mathbb{E}(Q_\tau^2) &= \frac{V_2^2 + 2V_1^2V_2 + (T-3)V_1^4}{T^3} + \frac{2\Re(V_4) + (T-2)V_3^2}{T^3} + \frac{2V_4' + (T-2)(V_3')^2}{T^3}, \text{ so that} \\ \text{Var}(Q_\tau) &= \frac{V_2^2 + 2V_1^2V_2 - 3V_1^4 + 2(\Re(V_4) + V_4') + (T-2)(V_3^2 + (V_3')^2)}{T^3}. \end{aligned}$$

For $k \neq \ell$ and both k, ℓ are non-zero, consider

$$\begin{aligned} \mathbb{E}(Q_k Q_\ell) &= \frac{1}{T^4} \sum_{t_1, s_1, t_2, s_2=1}^T \mathbb{E}(\mathbf{z}_{s_1}^* \Sigma_0 \mathbf{z}_{t_1} \mathbf{z}_{t_1-k}^* \Sigma_0 \mathbf{z}_{s_1-k} \mathbf{z}_{s_2}^* \Sigma_0 \mathbf{z}_{t_2} \mathbf{z}_{t_2-\ell}^* \Sigma_0 \mathbf{z}_{s_2-\ell}) \\ &= \frac{1}{T^4} \sum_{t, s=1}^T \mathbb{E}(\mathbf{z}_s^* \Sigma_0 \mathbf{z}_s \mathbf{z}_{s-k}^* \Sigma_0 \mathbf{z}_{s-k} \mathbf{z}_t^* \Sigma_0 \mathbf{z}_t \mathbf{z}_{t-\ell}^* \Sigma_0 \mathbf{z}_{t-\ell}) \\ &= \frac{4V_2V_1^2}{T^3} + \frac{(T^2 - 4T)V_1^4}{T^4}, \end{aligned}$$

where the last equality used the fact that when $t = s, t = s - k, s = t - \ell$ or $s - k = t - \ell$, the resulting expectation in the summation in the second equality is $V_2V_1^2$, otherwise it is V_1^4 . Hence

$$\text{Cov}(Q_k, Q_\ell) = \frac{4V_1^2(V_2 - V_1^2)}{T^3}.$$

With the above, we have

$$\begin{aligned} \text{Var}(G_q) &= \sum_{\tau=1}^q \text{Var}(Q_\tau) + \sum_{k \neq \ell} \text{Cov}(Q_k, Q_\ell) \\ &= q\text{Var}(Q_\tau) + q(q-1)\text{Cov}(Q_k, Q_\ell) \\ &= \frac{q(V_2 - V_1^2)^2 + 2q(\Re(V_4) + V_4') + q(T-2)(V_3^2 + (V_3')^2)}{T^3} + \frac{4q^2V_1^2(V_2 - V_1^2)}{T^3}. \end{aligned}$$

This completes the proof of the lemma. $\square$

APPENDIX B: CALCULATION OF THE LIMITING MOMENTS IN (4.5)-(4.7)

Consider first $f_1(x) = x$. As for the mean $\mathbb{E}X_{f_1}$, the first term in (4.3) with $f(z) = f_1(z) = z$ is equal to

$$\begin{aligned} & -\frac{1}{2\pi i} \oint_{\gamma} z \cdot \frac{c \int \frac{\underline{m}^3 t^2}{(1+t\underline{m})^3} dH(t)}{\left(1 - c \int \frac{t^2 \underline{m}^2}{(1+t\underline{m})^2} dH(t)\right)^2} dz \\ &= -\frac{1}{2\pi i} \oint_C \left(-\frac{1}{\underline{m}} + c \int \frac{t}{1+t\underline{m}} dH(t)\right) \cdot \frac{c \int \frac{\underline{m} t^2}{(1+t\underline{m})^3} dH(t)}{1 - c \int \frac{t^2 \underline{m}^2}{(1+t\underline{m})^2} dH(t)} d\underline{m} = 0. \end{aligned}$$

The second term of $\mathbb{E}X_{f_1}$ in (4.3) is equal to

$$\begin{aligned} & -\frac{\nu_4 - 3}{2\pi i} \oint_{\gamma} z \cdot \frac{c \int \frac{\underline{m}^3 t^2}{(1+t\underline{m})^3} dH(t)}{1 - c \int \frac{\underline{m}^2 t^2}{(1+t\underline{m})^2} dH(t)} dz \\ &= -\frac{\nu_4 - 3}{2\pi i} \oint_C \frac{-1 + c - c \int \frac{1}{1+t\underline{m}} dH(t)}{\underline{m}} \cdot \left(c \int \frac{t^2 \underline{m}}{(1+t\underline{m})^3} dH(t) \right) d\underline{m} = 0, \end{aligned}$$

Therefore, $\mathbb{E}X_{f_1} = 0$.

For the variance term $\text{Var}(X_{f_1})$, the first term in (4.4) with $f(z) = g(z) = f_1(z) = z$ is

$$\begin{aligned} I_1 &= -\frac{1}{2\pi^2} \iint \frac{z_1 z_2}{(\underline{m}(z_1) - \underline{m}(z_2))^2} d\underline{m}(z_1) d\underline{m}(z_2) \\ &= \frac{2}{2\pi i \cdot 2\pi i} \iint \frac{\left(-\frac{1}{\underline{m}_1} + c \int \frac{t}{1+t\underline{m}_1} dH(t)\right) \left(-\frac{1}{\underline{m}_2} + c \int \frac{t}{1+t\underline{m}_2} dH(t)\right)}{(\underline{m}_1 - \underline{m}_2)^2} d\underline{m}_1 d\underline{m}_2. \end{aligned}$$

We have

$$\begin{aligned} \frac{1}{2\pi i} \oint \frac{-\frac{1}{\underline{m}_1} + c \int \frac{t}{1+t\underline{m}_1} dH(t)}{(\underline{m}_1 - \underline{m}_2)^2} d\underline{m}_1 &= \frac{1}{2\pi i} \oint \frac{-1 + c - c \int \frac{1}{1+t\underline{m}_1} dH(t)}{\underline{m}_1 (\underline{m}_1 - \underline{m}_2)^2} d\underline{m}_1 \\ &= -\frac{-1 + c - c \int \frac{1}{1+t\underline{m}_1} dH(t)}{(\underline{m}_1 - \underline{m}_2)^2} \Big|_{\underline{m}_1=0} = \frac{1}{\underline{m}_2^2}. \end{aligned}$$

Therefore,

$$\begin{aligned} I_1 &= \frac{2}{2\pi i} \oint \frac{1}{\underline{m}_2^3} \left(-1 + c - c \int \frac{1}{1 + t\underline{m}_2} dH(t) \right) d\underline{m}_2 \\ &= -\frac{2}{2!} \left[-1 + c - c \int \frac{1}{1 + t\underline{m}_2} dH(t) \right]^{(2)} \Big|_{\underline{m}_2=0} = 2c \int t^2 dH(t). \end{aligned}$$

Here $[\cdot]^{(k)}|_{\underline{m}=0}$ represents the k -th order derivative of $[\cdot]$ with respect to $\underline{m}$ and then substitute with $\underline{m} = 0$.

The second term of $\text{Var}(X_{f_1})$ in (4.4) is

$$J_1 = -\frac{c(\nu_4 - 3)}{4\pi^2} \oint_{C_1} \oint_{C_2} z_1 z_2 \left[\int \frac{t}{(1 + t\underline{m}_1)^2} \cdot \frac{t}{(1 + t\underline{m}_2)^2} dH(t) \right] d\underline{m}_1 d\underline{m}_2.$$

We have

$$\begin{aligned} & \frac{1}{2\pi i} \oint_{C_1} z_1 \left[\int \frac{t}{(1 + t\underline{m}_1)^2} \cdot \frac{t}{(1 + t\underline{m}_2)^2} dH(t) \right] d\underline{m}_1 \\ &= - \left(-1 + c - c \int \frac{1}{1 + t\underline{m}_1} dH(t) \right) \cdot \left[\int \frac{t}{(1 + t\underline{m}_1)^2} \cdot \frac{t}{(1 + t\underline{m}_2)^2} dH(t) \right] \Big|_{\underline{m}_1=0} \\ &= \int \frac{t^2}{(1 + t\underline{m}_2)^2} dH(t). \end{aligned}$$

Therefore,

$$\begin{aligned} J_1 &= \frac{c(\nu_4 - 3)}{2\pi i} \oint_{C_2} z_2 \left(\int \frac{t^2}{(1 + t\underline{m}_2)^2} dH(t) \right) d\underline{m}_2 \\ &= -c(\nu_4 - 3) \left(-1 + c - c \int \frac{1}{1 + t\underline{m}_2} dH(t) \right) \left(\int \frac{t^2}{(1 + t\underline{m}_2)^2} dH(t) \right) \Big|_{\underline{m}_2=0} \\ &= c(\nu_4 - 3) \int t^2 dH(t) \end{aligned}$$

Finally,

$$\text{Var}(X_{f_1}) = I_1 + J_1 = c(\nu_4 - 1) \int t^2 dH(t) = c(\nu_4 - 1)s_2,$$

where $s_k = \int t^k dH(t)$. It follows that

$$\text{tr}(\widehat{\Sigma}_0) - \text{tr}(\Sigma_0) \xrightarrow{d} \mathcal{N}(0, c(\nu_4 - 1)s_2).$$

Similarly, we consider $tr(\widehat{\Sigma}_0^2)$ with the function $f_2(x) = x^2$. For the mean term $\mathbb{E}X_{f_2}$, the first term in (4.3) with $f(z) = f_2(z) = z^2$ is equal to

$$\begin{aligned} & -\frac{1}{2\pi i} \oint_{\gamma} z^2 \cdot \frac{c \int \frac{m^3 t^2}{(1+t\underline{m})^3} dH(t)}{\left(1 - c \int \frac{t^2 m^2}{(1+t\underline{m})^2} dH(t)\right)^2} dz \\ & = \left[\left(-1 + c - c \int \frac{1}{1+t\underline{m}} dH(t)\right)^2 \frac{c \int \frac{t^2}{(1+t\underline{m})^3} dH(t)}{1 - c \int \frac{t^2 m^2}{(1+t\underline{m})^2} dH(t)} \right]_{\underline{m}=0} \\ & = c \int t^2 dH(t) = cs_2. \end{aligned}$$

The second term is equal to

$$\begin{aligned} & -\frac{\nu_4 - 3}{2\pi i} \oint_{\gamma} z^2 \cdot \frac{c \int \frac{m^3 t^2}{(1+t\underline{m})^3} dH(t)}{1 - c \int \frac{m^2 t^2}{(1+t\underline{m})^2} dH(t)} dz \\ & = (\nu_4 - 3) \cdot \left[\left(-1 + c - c \int \frac{1}{1+t\underline{m}} dH(t)\right)^2 \cdot \left(c \int \frac{t^2}{(1+t\underline{m})^3} dH(t) \right) \right]_{\underline{m}=0} \\ & = c(\nu_4 - 3) \int t^2 dH(t) = c(\nu_4 - 3)s_2 \end{aligned}$$

Therefore, $\mathbb{E}X_{f_2} = c(\nu_4 - 2)s_2$.

For the variance term $\text{Var}(X_{f_2})$, the first term in (4.4) with $f(z) = g(z) = f_2(z) = z^2$ is equal to

$$\begin{aligned} I_2 &= -\frac{1}{2\pi^2} \oint \oint \frac{z_1^2 z_2^2}{(\underline{m}(z_1) - \underline{m}(z_2))^2} d\underline{m}(z_1) d\underline{m}(z_2) \\ &= \frac{2}{2\pi i \cdot 2\pi i} \oint \oint \frac{\left(-1 + c - c \int \frac{1}{1+t\underline{m}_1} dH(t)\right)^2 \left(-1 + c - c \int \frac{1}{1+t\underline{m}_2} dH(t)\right)^2}{\underline{m}_1^2 \underline{m}_2^2 (\underline{m}_1 - \underline{m}_2)^2} d\underline{m}_1 d\underline{m}_2. \end{aligned}$$

We have,

$$\begin{aligned} \frac{1}{2\pi i} \oint \frac{\left(-1 + c - c \int \frac{1}{1+t\underline{m}_1} dH(t)\right)^2}{\underline{m}_1^2 (\underline{m}_1 - \underline{m}_2)^2} d\underline{m}_1 &= - \left[\frac{\left(-1 + c - c \int \frac{1}{1+t\underline{m}_1} dH(t)\right)^2}{(\underline{m}_1 - \underline{m}_2)^2} \right]_{\underline{m}_1=0}^{(1)} \\ &= \frac{2c}{\underline{m}_2^2} \int t dH(t) - \frac{2}{\underline{m}_2^3}. \end{aligned}$$

Therefore,

$$\begin{aligned}
I_2 &= \frac{2}{2\pi i} \oint \left(\frac{2c}{\underline{m}_2^2} \int t \, dH(t) - \frac{2}{\underline{m}_2^3} \right) \cdot \frac{\left(-1 + c - c \int \frac{1}{1+\underline{t}\underline{m}_2} dH(t) \right)^2}{\underline{m}_2^2} d\underline{m}_2 \\
&= -\frac{4cs_1}{3!} \left[\left(-1 + c - c \int \frac{1}{1+\underline{t}\underline{m}_2} dH(t) \right)^2 \right]^{(3)} \Big|_{\underline{m}_2=0} + \frac{4}{4!} \left[\left(-1 + c - c \int \frac{1}{1+\underline{t}\underline{m}_2} dH(t) \right)^2 \right]^{(4)} \Big|_{\underline{m}_2=0} \\
&= 8c^3 s_1^2 s_2 + 16c^2 s_1 s_3 + 4c^2 s_2^2 + 8cs_4.
\end{aligned}$$

The second term of $\text{Var}(X_{f_2})$ in (4.4) is equal to

$$J_2 = -\frac{c(\nu_4 - 3)}{4\pi^2} \oint_{C_1} \oint_{C_2} z_1^2 z_2^2 \left[\int \frac{t}{(1+\underline{t}\underline{m}_1)^2} \cdot \frac{t}{(1+\underline{t}\underline{m}_2)^2} dH(t) \right] d\underline{m}_1 d\underline{m}_2.$$

We have

$$\begin{aligned}
&\frac{1}{2\pi i} \oint_{C_1} z_1^2 \left[\int \frac{t}{(1+\underline{t}\underline{m}_1)^2} \cdot \frac{t}{(1+\underline{t}\underline{m}_2)^2} dH(t) \right] d\underline{m}_1 \\
&= -\left[\left(-1 + c - c \int \frac{1}{1+\underline{t}\underline{m}_1} dH(t) \right)^2 \cdot \left[\int \frac{t}{(1+\underline{t}\underline{m}_1)^2} \cdot \frac{t}{(1+\underline{t}\underline{m}_2)^2} dH(t) \right] \right]^{(1)} \Big|_{\underline{m}_1=0} \\
&= 2c \left(\int t \, dH(t) \right) \left(\int \frac{t^2}{(1+\underline{t}\underline{m}_2)^2} dH(t) \right) + 2 \int \frac{t^3}{(1+\underline{t}\underline{m}_2)^2} dH(t).
\end{aligned}$$

Therefore,

$$\begin{aligned}
J_2 &= \frac{c(\nu_4 - 3)}{2\pi i} \oint_{C_2} z_2^2 \left(2cs_1 \int \frac{t^2}{(1+\underline{t}\underline{m}_2)^2} dH(t) + 2 \int \frac{t^3}{(1+\underline{t}\underline{m}_2)^2} dH(t) \right) d\underline{m}_2 \\
&= \frac{c(\nu_4 - 3)}{2\pi i} \oint_{C_2} \left(\frac{-1 + c - c \int \frac{1}{1+\underline{t}\underline{m}_2} dH(t)}{\underline{m}_2} \right)^2 \left(2cs_1 \int \frac{t^2}{(1+\underline{t}\underline{m}_2)^2} dH(t) + 2 \int \frac{t^3}{(1+\underline{t}\underline{m}_2)^2} dH(t) \right) d\underline{m}_2 \\
&= c(\nu_4 - 3) (4c^2 s_1^2 s_2 + 8cs_1 s_3 + 4s_4).
\end{aligned}$$

Finally,

$$\text{Var}(X_{f_2}) = I_2 + J_2 = 4c^2 s_2^2 + (\nu_4 - 1)(4c^3 s_1^2 s_2 + 8c^2 s_1 s_3 + 4cs_4).$$

It follows in particular that

$$\text{Tr}(\widehat{\Sigma}_0^2) - \text{Tr}(\Sigma_0^2) - \frac{c}{p} (tr(\Sigma_0))^2 \xrightarrow{d} \mathcal{N}\left(cs_2(\nu_4 - 2), 4c^2 s_2^2 + (\nu_4 - 1)(4c^3 s_1^2 s_2 + 8c^2 s_1 s_3 + 4cs_4)\right).$$

Last but not least, we consider the covariance of X_{f_1} and X_{f_2} . The first term in (4.4) with $f(z) = f_1(z) = z$ and $g(z) = f_2(z) = z^2$ is

$$\begin{aligned} -\frac{1}{2\pi^2} \oint\!\!\!\oint \frac{z_1 z_2^2}{(\underline{m}_1 - \underline{m}_2)^2} d\underline{m}_1 d\underline{m}_2 &= \frac{2}{2\pi i} \oint \frac{1}{\underline{m}_2^4} \left(-1 + c - c \int \frac{1}{1 + t\underline{m}_2} dH(t) \right)^2 d\underline{m}_2 \\ &= -\frac{2}{3!} \left[\left(-1 + c - c \int \frac{1}{1 + t\underline{m}_2} dH(t) \right)^2 \right]^{(3)} \Big|_{\underline{m}_2=0} \\ &= 4c^2 s_1 s_2 + 4cs_3. \end{aligned}$$

Here we have used the fact

$$\frac{1}{2\pi i} \oint \frac{z_1}{(\underline{m}_1 - \underline{m}_2)^2} d\underline{m}_1 = \frac{1}{\underline{m}_2^2}.$$

As for the second term of $\text{Cov}(X_{f_1}, X_{f_2})$ in (4.4), we have

$$\begin{aligned} &\frac{1}{2\pi i} \oint_{C_1} z_1 \left[\int \frac{t}{(1 + t\underline{m}_1)^2} \cdot \frac{t}{(1 + t\underline{m}_2)^2} dH(t) \right] d\underline{m}_1 \\ &= \frac{1}{2\pi i} \oint_{C_1} \frac{-1 + c - c \int \frac{1}{1 + t\underline{m}_1} dH(t)}{\underline{m}_1} \cdot \left[\int \frac{t}{(1 + t\underline{m}_1)^2} \cdot \frac{t}{(1 + t\underline{m}_2)^2} dH(t) \right] d\underline{m}_1 \\ &= \int \frac{t^2}{(1 + t\underline{m}_2)^2} dH(t). \end{aligned}$$

Then that second term is equal to

$$\begin{aligned} &\frac{c(v_4 - 3)}{2\pi i} \oint_{C_2} z_2^2 \left(\int \frac{t^2}{(1 + t\underline{m}_2)^2} dH(t) \right) d\underline{m}_2 \\ &= -c(v_4 - 3) \left[\left(-1 + c - c \int \frac{1}{1 + t\underline{m}_2} dH(t) \right)^2 \left(\int \frac{t^2}{(1 + t\underline{m}_2)^2} dH(t) \right) \right]^{(1)} \Big|_{\underline{m}_2=0} \\ &= c(v_4 - 3) (2cs_1 s_2 + 2s_3). \end{aligned}$$

Finally the limiting covariance is

$$\text{Cov}(X_{f_1}, X_{f_2}) = (v_4 - 1)(2c^2 s_1 s_2 + 2cs_3).$$

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In conclusion, we have established that

$$\begin{pmatrix} \text{Tr}(\widehat{\Sigma}_0) - \text{Tr}(\Sigma_0) \\ \text{Tr}(\widehat{\Sigma}_0^2) - \text{Tr}(\Sigma_0^2) - \frac{1}{T}(\text{Tr}(\Sigma_0))^2 \end{pmatrix} \xrightarrow{d} \mathcal{N}_2 \left(\begin{pmatrix} 0 \\ cs_2(\nu_4 - 2) \end{pmatrix}, \begin{pmatrix} (\nu_4 - 1)cs_2 & (\nu_4 - 1)(2c^2s_1s_2 + 2cs_3) \\ (\nu_4 - 1)(2c^2s_1s_2 + 2cs_3) & 4c^2s_2^2 + (\nu_4 - 1)(4c^3s_1^2s_2 + 8c^2s_1s_3 + 4cs_4) \end{pmatrix} \right).$$

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